

WEYL–CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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ABSTRACT.

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1. INTRODUCTION

Theorem 1.1. *Let R be a normalised root system in $\mathbb{R}^N$, $N \geq 2$, and let G be the corresponding finite reflection group. Let $k : R \rightarrow \mathbb{R}_+$ be G -invariant multiplicity function. Let $R_{\text{Dunkl},j}$ be the j -th Dunkl-Riesz transform corresponding to R and k . Let w be the Dunkl weight corresponding to R and k . Let $f \in L_1^{\text{loc}}(\mathbb{R}^N)$.*

- (i) *If $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, then $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Furthermore,*

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{R,k} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

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(ii) If $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, then $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. Furthermore,

$$\|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \geq c_{R,k}^{-1} \|f\|_{\dot{W}^{1,N}(\mathbb{R}^N)};$$

2. PRELIMINARIES

2.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $\|\cdot\|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane orthogonal to a normalised vector v is defined by

$$\sigma_v x := x - 2\langle x, v \rangle v.$$

A finite set $R \subset \mathbb{R}^N \setminus \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalised reduced root system if $\|v\|^2 = 1$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v \in R}$ is called the Weyl group (reflection group) of the root system R . Let $\kappa : R \rightarrow [0, \infty)$ be a multiplicity function, which is a G -invariant function fixed throughout the paper.

Define $d(x, y) = \min_{g \in G} |x - gy|$. Recall from [8, Chapter VII, proof of Theorem 2.12] that $d(x, y) = |x - y|$ iff x and y belong to the same chamber.

Define the Dunkl measure dw on $\mathbb{R}^N$ by

$$dw(x) := \prod_{v \in R} |\langle x, v \rangle|^{\kappa(v)} dx,$$

where dx stands for the Lebesgue measure on $\mathbb{R}^N$. Let $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Let $\mathbf{N} := N + \sum_{v \in R} \kappa(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, t, r > 0.$$

Furthermore,

$$(1) \quad w(B(x, r)) \sim r^N \prod_{v \in R} (|\langle x, v \rangle| + r)^{\kappa(v)},$$

where the implicit constant only depends on N, R and κ . It follows from (1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$(2) \quad C^{-1} \left(\frac{r_2}{r_1} \right)^N \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2,$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v \in R} \frac{\kappa(v)}{2} \langle v, \xi \rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x \rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ . Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} = - \sum_{j=1}^N T_{e_j}^2.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} = T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}$, $j = 1, 2, \dots, N$.

2.2. Weyl chamber as simplicial cone.

Definition 2.1. A connected component of the set

$$\{x \in \mathbb{R}^N : \langle x, v \rangle \neq 0 \text{ for all } v \in R\}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber.

The following assertion can be found in e.g. Theorem 2 on p.166 in [1].

Theorem 2.2. Let Γ be a chamber. There exists a linear isomorphism $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ such that $\Phi(\Gamma) = R_+^{N_1} \times \mathbb{R}^{N_2}$ (with $N_1 + N_2 = N$).

2.3. System of dyadic cubes.

Notation 2.3. Let

$$\mathbb{D} = \bigcup_{k \in \mathbb{Z}} \mathbb{D}_k$$

be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube in $\mathbb{D}_k$ is of the shape

$$\frac{m}{2^k} + [0, 2^{-k})^N, \quad m \in \mathbb{Z}^N.$$

For every dyadic cube $Q \in \mathbb{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.

2.4. Nearly weakly orthonormal sequences. Rochberg and Semmes [13] introduced the notion of *nearly weakly orthonormal* (NWO) sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [5, 10, 12, 3, 4, 11, 13, 15, 14]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [10, Section 13] for a complete proof).

Definition 2.4 and Lemma 2.5 as stated below are my understanding of Hytonen's Corollary 8.2 and Lemma 8.7. Please check whether this understanding is the correct one!

Definition 2.4. Let $\mathbf{w} \in A_2(\mathbb{R}^N)$. Let $c > 0$ and $2 < r \leq \infty$. A sequence of functions $\{e_Q\}_{Q \in \mathbb{D}}$ is called (r, c) -NWO if

- (i) $\text{supp}(e_Q) \subseteq cQ$ for every $Q \in \mathbb{D}$;
- (ii) $\|e_Q\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} \leq \mathbf{w}(Q)^{\frac{1}{r} - \frac{1}{2}}$ for every $Q \in \mathbb{D}$;

Lemma 2.5. Let $\mathbf{w} \in A_2(\mathbb{R}^N)$. Let $0 < p < \infty$. Suppose that A is an integral operator on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$ with integral kernel

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

where $\{e_Q\}_{Q \in \mathbb{D}}$ and $\{f_Q\}_{Q \in \mathbb{D}}$ are (r_1, c_1) -NWO and (r_2, c_2) -NWO sequences, respectively. We have

$$\|A\|_{\mathcal{L}_{p,\infty}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))} \leq C_{w,r_1,c_1,r_2,c_2,p} \|\{\lambda_Q\}_{Q \in \mathbb{D}}\|_{p,\infty},$$

$$\begin{aligned} \text{dist}_{\mathcal{L}_{p,\infty}(L_2(\Gamma, \mathbf{w}(x)dx))} \left(A, (\mathcal{L}_{p,\infty})_0(L_2(\mathbb{R}^N, \mathbf{w}(x)dx)) \right) &\leq \\ &\leq C_{\mathbf{w}, r_1, c_1, r_2, c_2, p} \text{dist}_{l_{p,\infty}(\mathbb{D})} \left(\{\lambda_Q\}_{Q \in \mathbb{D}}, (l_{p,\infty})_0(\mathbb{D}) \right). \end{aligned}$$

Proof. The assertion follows from Corollary 8.2 and Lemma 8.7 in [10]. $\square$

Lemma 2.6. *Let $w \in A_2(\mathbb{R}^N)$. Let $1 < p < \infty$. Suppose that A is an integral operator on $L_2(\mathbb{R}^N, \mathbf{w}(x)dx)$ with integral kernel*

$$(x, y) \rightarrow \sum_{Q \in \mathbb{D}} \lambda_Q e_Q(x) f_Q(y), \quad x, y \in \mathbb{R}^N,$$

where $\{e_Q\}_{Q \in \mathbb{D}}$ and $\{f_Q\}_{Q \in \mathbb{D}}$ are (r_1, c_1) -NWO and (r_2, c_2) -NWO sequences, respectively. We have

$$\left\| \left\{ \langle A e_Q, f_Q \rangle_{L_2(\mathbb{R}^N, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{l_{p,q}(\mathbb{D})} \leq c_{\mathbf{w}, r_1, c_1, r_2, c_2, p, q} \|A\|_{\mathcal{L}_{p,q}(L_2(\mathbb{R}^N, \mathbf{w}(x)dx))}.$$

Proof. The assertion follows from Corollary 8.2 and Lemma 8.7 in [10]. $\square$

3. CHAMBER IS A DOUBLING METRIC MEASURE SPACE

Theorem 3.1. *Let Γ be a chamber. For all $x \in \Gamma$ and for all $r > 0$, we have*

$$w(B(x, r) \cap \Gamma) \geq c_\kappa w(B(x, r)).$$

In particular, $(\Gamma, |\cdot|, w(x)dx)$ is a doubling metric measure space.

Notation 3.2. *Let $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$. Set*

$$w_\theta(x) = \prod_{A \subset \{1, \dots, N_1\}} \left(\sum_{k \in A} x_k \right)^{\theta_A}, \quad x \in \mathbb{R}_+^{N_1}.$$

Lemma 3.3. *For every $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ and for every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$, we have*

$$\int_{B(x, r)} w_\theta(y_1, \dots, y_{N_1}) dy \leq c_{N, \theta}^{(1)} r^N w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

Proof. Obviously,

$$B(x, r) \subset \left(\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r).$$

Thus,

$$\begin{aligned} &\int_{B(x, r)} w_\theta(y_1, \dots, y_{N_1}) dy \leq \\ &\leq \int_{\left(\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r] \right) \times B((x_{N_1+1}, \dots, x_N), r)} w_\theta(y_1, \dots, y_{N_1}) dy = \\ &= \int_{\times_{k=1}^{N_1} [(x_k - r)_+, x_k + r]} w_\theta(u) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\ &\leq \int_{\times_{k=1}^{N_1} [0, x_k + r]} w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) du \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} = \\ &= w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times \prod_{k=1}^{N_1} (x_k + r - (x_k - r)_+) \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2} \leq \\ &\leq w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times (2r)^{N_1} \times \text{Vol}(\mathbb{B}_{N_2}) r^{N_2}. \end{aligned}$$

$\square$

Lemma 3.4. *For every $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ and for every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$, we have*

$$\int_{B(x,r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq c_{N,\theta}^{(2)} r^N w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}).$$

Proof. Obviously,

$$x + [\frac{r}{2N}, \frac{r}{N}]^N \subset B(x, r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}).$$

It follows that

$$\begin{aligned} & \int_{B(x,r) \cap (\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2})} w_\theta(y_1, \dots, y_{N_1}) dy \geq \\ & \geq \int_{x + [\frac{r}{2N}, \frac{r}{N}]^N} w_\theta(y_1, \dots, y_{N_1}) dy = \\ & = \int_{(x_1, \dots, x_{N_1}) + [\frac{r}{2N}, \frac{r}{N}]^{N_1}} w_\theta(u) du \times (\frac{r}{2N})^{N_2} \geq \\ & \geq \int_{(x_1, \dots, x_{N_1}) + [\frac{r}{2N}, \frac{r}{N}]^{N_1}} w_\theta((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) du \times (\frac{r}{2N})^{N_2} = \\ & = w_\theta((x_1, \dots, x_{N_1}) + \frac{r}{2N} \mathbf{1}_{N_1}) \times (\frac{r}{2N})^N \geq \\ & \geq w_\theta((x_1, \dots, x_{N_1}) + r \mathbf{1}_{N_1}) \times (\frac{r}{2N})^N \times (2N)^{-\sum_{A \subset \{1, \dots, N_1\}} \theta_A}. \end{aligned}$$

□

Lemma 3.5. *There exists $\theta : 2^{N_1} \rightarrow \mathbb{R}_+$ such that*

$$c_\kappa^{-1} w_\theta(x_1, \dots, x_{N_1}) \leq w(\Phi^{-1}(x)) \leq c_\kappa w_\theta(x_1, \dots, x_{N_1}), \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Proof. For every $x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$ and for every $v \in R$, we have

$$\langle \Phi^{-1}(x), v \rangle = \langle x, (\Phi^{-1})^*(v) \rangle.$$

The latter expression has the same sign (positive or negative) on the whole cone $\mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. Hence,

$$((\Phi^{-1})^*(v))_k = \begin{cases} \text{something positive,} & 1 \leq k \leq N_1 \\ 0, & k > N_1 \end{cases}$$

Let A_v be the set of those $1 \leq k \leq N_1$ with $((\Phi^{-1})^*(v))_l$ being strictly positive. We, obviously, have

$$c_{\Phi,v}^{-1} \sum_{k \in A_v} x_k \leq |\langle \Phi^{-1}(x), v \rangle| \leq c_{\Phi,v} \sum_{k \in A_v} x_k, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

Thus,

$$\prod_{v \in R} c_{\Phi,v}^{-\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)} \leq w(x) \leq \prod_{v \in R} c_{\Phi,v}^{\kappa(v)} \left(\sum_{k \in A_v} x_k \right)^{\kappa(v)}, \quad x \in \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}.$$

This completes the proof. □

Proof of Theorem 3.1. The first assertion follows by combining Lemmas 3.3, 3.4 and 3.5. The second assertion is an obvious consequence of the first one. □

4. ROCHBERG-SEMMES BASED ESTIMATE

Theorem 4.1. *Let $N \geq 2$, $1 \leq N_1 \leq N$ and $N_2 = N - N_1$. Let $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. Let $\mathbf{w} \in A_2(\Omega)$ be such that*

$$\sup_{x \in Q} \mathbf{w}(x) \leq c_w \frac{\mathbf{w}(Q)}{m(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

Suppose that $K : \Omega \times \Omega \rightarrow \mathbb{C}$ satisfies

$$|\partial_x^\alpha \partial_y^\beta K|(x, y) \leq c_{K, \alpha, \beta} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{\mathbf{w}(B(x, |x - y|))}, \quad x, y \in \Omega, \quad \alpha, \beta \in \mathbb{Z}_+^N.$$

Let V be an integral operator on $L_2(\Omega, \mathbf{w}(x)dx)$ with integral kernel K . Suppose $N > 2s$. For every $f \in C_c^\infty(\mathbb{R}^N)$, we have

$$\|M_f V\|_{\mathcal{L}_{N, \infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq c_{K, \mathbf{w}} \|f\|_{L_N(\Omega)}.$$

Lemma 4.2. *There exists a smooth function ψ supported on $[-1, 1]^N \setminus (-\frac{1}{4}, \frac{1}{4})^N$ such that*

$$1 = \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x - y}{\ell(Q)}\right).$$

Proof. Choose $\rho \in C_c^\infty(\mathbb{R}^N)$ such that $\rho = 1$ on $(-\frac{1}{2}, \frac{1}{2})^N$ and $\rho = 0$ outside $(-1, 1)^N$. Set $\psi(z) = \rho(z) - \rho(2z)$. Clearly,

$$\text{supp}(\psi) \subset \left\{ z \in \mathbb{R}^N : \frac{1}{4} \leq |z|_\infty \leq 1 \right\}.$$

For every $z \neq 0$, we have

$$(3) \quad \sum_{j \in \mathbb{Z}} \psi(2^j z) = 1.$$

For every x , there exists a unique $Q \in \mathbb{D}_j$ containing x . For this Q , we have $\ell(Q) = 2^{-j}$. We have

$$\begin{aligned} 1 &= \sum_{j \in \mathbb{Z}} \psi(2^j(x - y)) = \sum_{j \in \mathbb{Z}} \left(\sum_{Q \in \mathbb{D}_j} \chi_Q(x) \right) \psi(2^j(x - y)) = \\ &= \sum_{j \in \mathbb{Z}} \sum_{Q \in \mathbb{D}_j} \chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right) = \sum_{Q \in \mathbb{D}} \chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right). \end{aligned}$$

If $x \in Q$ and $\psi(\frac{x-y}{\ell(Q)}) \neq 0$, then

$$x \in Q, \quad |x - y|_\infty \leq \ell(Q).$$

We obviously have $|x - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$ and $|x - y|_\infty \leq \ell(Q)$. Thus, $|y - c_Q|_\infty \leq \frac{3}{2}\ell(Q)$. In other words, $y \in 3Q$. Thus,

$$\chi_Q(x) \psi\left(\frac{x - y}{\ell(Q)}\right) = \chi_Q(x) \chi_{3Q}(y) \psi\left(\frac{x - y}{\ell(Q)}\right).$$

This completes the proof. $\square$

Lemma 4.3. *Let $r > 2$. In the setting of Theorem 4.1, we have*

$$f(x)K(x, y) = \sum_{k, l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k, l, Q}(x) f_{k, l, Q}(y),$$

where

(i) $e_{k,l,Q}$ is supported in Q and $f_{k,l,Q}$ is supported in $3Q$ for every $Q \in \mathbb{D}$ with $Q \subset \Omega$;

(ii) for every $k, l \in \mathbb{Z}^N$ and for every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have

$$\|e_{k,l,Q}\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} = \|f\|_{L_r(Q, \mathbf{w}(x)dx)};$$

(iii) for every $k, l \in \mathbb{Z}^N$ and for every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have

$$\|f_{k,l,Q}\|_{L_\infty(\mathbb{R}^N)} \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)};$$

Proof. For every $x, y \in \Omega$, we have

$$\begin{aligned} K(x, y) &= \sum_{Q \in \mathbb{D}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x-y}{\ell(Q)}\right) = \\ &= \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \chi_Q(x) \chi_{3Q}(y) K(x, y) \psi\left(\frac{x-y}{\ell(Q)}\right) = \end{aligned}$$

For every $Q \in \mathbb{D}$ with $3Q \subset \Omega$, define the map

$$K_Q : \left[-\frac{1}{2}, \frac{1}{2}\right]^N \times \left[-\frac{3}{2}, \frac{3}{2}\right]^N \rightarrow \mathbb{C}$$

by setting

$$K_Q(a, b) = K(c_Q + a\ell(Q), c_Q + b\ell(Q))\psi(a - b), \quad a \in \left[-\frac{1}{2}, \frac{1}{2}\right]^N, \quad b \in \left[-\frac{3}{2}, \frac{3}{2}\right]^N.$$

If $Q \in \mathbb{D}$ is such that $3Q \not\subset \Omega$, then the definition of K_Q is the same, a also runs through $[-\frac{1}{2}, \frac{1}{2}]^N$, but the parallelepiped b runs through is slightly smaller. **to co-authors: if you wish, specify the domain of b also.** In what follows, we lighten the notations by pretending that b is always defined on $[-\frac{3}{2}, \frac{3}{2}]^N$.

It follows from the assumption on K that

$$\|K_Q\|_{C^{2N+2}([- \frac{3}{2}, \frac{3}{2}]^N \times [- \frac{3}{2}, \frac{3}{2}]^N)} \leq \frac{c'_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

Hence, there exists an extension $K_Q : \mathbb{T}^N \times \mathbb{T}^N \rightarrow \mathbb{C}$ such that

$$\|K_Q\|_{C^{2N+2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c''_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

In particular,

$$\|K_Q\|_{W^{2N+2,2}(\mathbb{T}^N \times \mathbb{T}^N)} \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}.$$

We may, therefore, write

$$K_Q(a, b) = \sum_{k,l \in \mathbb{Z}^N} \text{coef}_{k,l,Q} e^{i\langle k,a \rangle} e^{i\langle l,b \rangle}, \quad a, b \in \mathbb{T}^N,$$

where

$$\sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{2N+2} |\text{coef}_{k,l,Q}|^2 \leq \left(\frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)} \right)^2.$$

In particular,

$$(1 + |k|^2 + |l|^2)^{N+1} \cdot |\text{coef}_{k,l,Q}| \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}, \quad k, l \in \mathbb{Z}^N.$$

We now set

$$e_{k,l,Q}(x) = f(x)\chi_Q(x)e^{i\langle k, \frac{x-c_Q}{\ell(Q)} \rangle},$$

$$f_{k,l,Q}(y) = (1 + |k|^2 + |l|^2)^{N+1} \text{coef}_{k,l,Q} \cdot \chi_{3Q}(y)e^{i\langle l, \frac{y-c_Q}{\ell(Q)} \rangle}.$$

This allows us to write

$$f(x)K(x, y) = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k,l,Q}(x) f_{k,l,Q}(y).$$

Obviously,

$$\begin{aligned} \text{supp}(e_{k,l,Q}) &\subset Q, \quad \text{supp}(f_{k,l,Q}) \subset 3Q, \\ \|e_{k,l,Q}\|_{L_r(\mathbb{R}^N, \mathbf{w}(x)dx)} &= \|f\|_{L_r(Q, \mathbf{w}(x)dx)}, \\ \|f_{k,l,Q}\|_{L_\infty(\mathbb{R}^N)} &= (1 + |k|^2 + |l|^2)^{N+1} |\text{coef}_{k,l,Q}| \leq \frac{c_{K,\theta} \cdot \ell(Q)}{\mathbf{w}(Q)}. \end{aligned}$$

□

Lemma 4.4. *Let $r > 2$. In the setting of Theorem 4.1, we have*

$$\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq c_{K,\mathbf{w},r} \left\| \left\{ m(Q)^{\frac{1}{N}-\frac{1}{r}} \|f\|_{L_r(Q)} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \right\|_{N,\infty}.$$

Proof. Let $A_{k,l}$ be the integral operator with integral kernel given by the formula

$$(x, y) \rightarrow \sum_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} e_{k,l,Q}(x) f_{k,l,Q}(y), \quad x, y \in \Omega, \quad k, l \in \mathbb{Z}^N.$$

By Lemma 4.3, we have

$$M_f V = \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} A_{k,l}.$$

By the triangle inequality,

$$\begin{aligned} &\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, w_\theta(x)dx))} \leq \\ &\leq \frac{N}{N-1} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, w_\theta(x)dx))}. \end{aligned}$$

By Lemma 4.3, the sequences

$$\left\{ \mathbf{w}(Q)^{\frac{1}{r}-\frac{1}{2}} \frac{e_{k,l,Q}}{\|f\|_{L_r(Q, \mathbf{w}(x)dx)}} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}}, \quad \left\{ \frac{c_{K,\theta}^{-1} \mathbf{w}(Q)^{\frac{1}{2}}}{\ell(Q)} f_{k,l,Q} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}}$$

are $(r, 1)$ -NWO and, respectively, $(\infty, 3)$ -NWO in the sense of Definition 2.4. Using Lemma 2.5, we conclude that

$$\begin{aligned} &\|A_{k,l}\|_{\mathcal{L}_{N,\infty}(L_2(\Omega, \mathbf{w}(x)dx))} \leq \\ &\leq c_{\theta,r,1,\infty,3,N} c_{K,\theta} \left\| \left\{ \ell(Q) \mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \right\}_{\substack{Q \in \mathbb{D} \\ Q \subset \Omega}} \right\|_{N,\infty}. \end{aligned}$$

Therefore,

$$\begin{aligned} &\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}_+^N, \mathbf{w}(x)dx))} \leq \\ &\leq \frac{N}{N-1} c_{\theta,r,1,\infty,3,N} c_{K,\theta} \sum_{k,l \in \mathbb{Z}^N} (1 + |k|^2 + |l|^2)^{-N-1} \times \end{aligned}$$

$$\times \left\| \left\{ \ell(Q) \mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \right\}_{Q \in \mathbb{D}} \right\|_{N, \infty}.$$

Obviously,

$$\mathbf{w}(Q)^{-\frac{1}{r}} \|f\|_{L_r(Q, \mathbf{w}(x)dx)} \leq \left(\frac{1}{\mathbf{w}(Q)} \cdot \sup_{x \in Q} \mathbf{w}(x) \right)^{\frac{1}{r}} \times \|f\|_{L_r(Q)}, \quad Q \in \mathbb{D}, \quad Q \subset \Omega.$$

Combining the last two estimates, we complete the proof. $\square$

Proof of Theorem 4.1. Take $r \in (2, N)$. The assertion follows by combining Lemmas 4.4 and A.1. $\square$

5. PROOF OF SPECIAL CBIKEL ESTIMATE IN DUNKL SETTING

Lemma 5.1. *For every pair of multi-indices $\alpha, \beta \in \mathbb{Z}_+^N$, for every $t > 0$, we have*

$$|(\partial_x^\alpha \partial_y^\beta h_t)(x, y)| \leq c_{\kappa, \alpha, \beta} t^{1 - \frac{|\alpha|_1 + |\beta|_1}{2}} d(x, y)^{-2} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2).$$

Proof. This is (slightly weakened form of) Theorem 3.1 in [2]. $\square$

Lemma 5.2. *Suppose $s + N > 2$. We have*

$$\int_0^\infty t^{-\frac{s}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt \leq c_{\kappa, s} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s}.$$

Proof. By (2),

$$w(B(x, t^{\frac{1}{2}}))^{-1} \leq c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \begin{cases} \left(\frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N, & 0 < t \leq d(x, y)^2, \\ \left(\frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N, & t \geq d(x, y)^2. \end{cases}$$

Therefore,

$$\begin{aligned} & \int_0^\infty t^{-\frac{s}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt \leq \\ & \leq c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \int_0^{d(x, y)^2} t^{-\frac{s}{2}} \left(\frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt + \\ & \quad + c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times \int_{d(x, y)^2}^\infty t^{-\frac{s}{2}} \left(\frac{d(x, y)}{t^{\frac{1}{2}}} \right)^N dt = \\ & = c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s} \times \int_0^1 t^{-\frac{s+N}{2}} \exp(-\frac{c_\kappa}{t}) dt + \\ & \quad + c_\kappa^{(1)} w(B(x, d(x, y)))^{-1} \times d(x, y)^{2-s} \times \int_1^\infty t^{-\frac{s+N}{2}} dt. \end{aligned}$$

$\square$

The following lemma establishes derivative estimates for the integral kernels of Dunkl-Riesz transforms.

Lemma 5.3. *For every pair of multi-indices α, β , for every $1 \leq j \leq N$ with $N + |\alpha|_1 + |\beta|_1 > 3$, we have*

$$|(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| \leq c_{\kappa, \alpha, \beta} \frac{d(x, y)^{1 - |\alpha|_1 - |\beta|_1}}{w(B(x, d(x, y)))}, \quad x, y \in \mathbb{R}^N.$$

an issue here is that for $\alpha = \beta = 0$, this imposes the condition $N \geq 4$. it would be good to prove something like this under the assumption $N \geq 3$.

Proof. We have

$$\Delta_{\text{Dunkl}}^{-\frac{1}{2}} = \pi^{-\frac{1}{2}} \int_0^\infty e^{-t\Delta_{\text{Dunkl}}} \frac{dt}{t^{\frac{1}{2}}}.$$

Hence,

$$K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}}(x, y) = \pi^{-\frac{1}{2}} \int_0^\infty h_t(x, y) \frac{dt}{t^{\frac{1}{2}}}.$$

Using Lemma 5.1, we estimate

$$\begin{aligned} & |(\partial_x^\alpha \partial_y^\beta K_{R_{\text{Dunkl},j}})(x, y)| \leq \\ & \leq \frac{c_{\kappa, \alpha, \beta}}{\pi^{\frac{1}{2}} d(x, y)^2} \int_0^\infty t^{-\frac{|\alpha|_1 + |\beta|_1 - 1}{2}} w(B(x, t^{\frac{1}{2}}))^{-1} \exp(-\frac{c_\kappa}{t} d(x, y)^2) dt. \end{aligned}$$

Applying Lemma 5.2, we conclude

$$\begin{aligned} & |(\partial_x^\alpha \partial_y^\beta K_{\Delta_{\text{Dunkl}}^{-\frac{1}{2}}})(x, y)| \leq \\ & \leq \pi^{-\frac{1}{2}} c_{\kappa, \alpha, \beta} \times c_{\kappa, |\alpha|_1 + |\beta|_1 - 1} w(B(x, d(x, y)))^{-1} \times d(x, y)^{1 - |\alpha|_1 - |\beta|_1}. \end{aligned}$$

The assertion follows immediately. $\square$

Lemma 5.4. *For every $Q \in \mathbb{D}$ with $Q \subset \Omega$, we have*

$$\sup_{x \in Q} w_\theta(x) \leq c_\theta \frac{w_\theta(Q)}{m(Q)}.$$

Proof. We have

$$w(Q) \geq c_\theta \prod_{A \subset \{1, \dots, N\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

On the other hand, if $x \in Q$, then $|x - c_Q|_\infty \leq \frac{1}{2} \ell(Q)$ and, therefore,

$$\sum_{k \in A} x_k \leq \frac{|A|}{2} \ell_Q + \sum_{k \in A} (c_Q)_k \leq N(\ell_Q + \sum_{k \in A} (c_Q)_k).$$

Thus,

$$w(x) = \prod_{A \subset \{1, \dots, N\}} (\sum_{k \in A} x_k)^{\theta_A} \leq N^{\sum_{A \subset \{1, \dots, N\}} \theta_A} \prod_{A \subset \{1, \dots, N\}} (\ell_Q + \sum_{k \in A} (c_Q)_k)^{\theta_A}.$$

This completes the proof. $\square$

Theorem 5.5. *Let $N \geq 4$. We have*

$$\|M_f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{s}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_\kappa \|f\|_{L_N(\mathbb{R}^N)}.$$

Proof. It suffices to estimate, for every chamber Γ and for every $g \in G$, the operator $M_{\chi_\Gamma \cdot f} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} M_{\chi_{g\Gamma}}$. In other words, it suffices to estimate an operator with integral kernel

$$f(x) K_{\text{Dunkl}^{-\frac{1}{2}}}(x, y), \quad x, y \in \Gamma, \quad y \in g\Gamma,$$

on the Hilbert space $L_2(\mathbb{R}^N, w(x)dx)$. In other words, it suffices to estimate an operator with integral kernel

$$f(x) K_{\text{Dunkl}^{-\frac{1}{2}}}(x, gy), \quad x, y \in \Gamma,$$

on the Hilbert space $L_2(\Gamma, w(x)dx)$.

Let $N_1 = \dim(\text{span}(R))$, $N_2 = N - N_1$ and let $\Omega = \mathbb{R}_+^{N_1} \times \mathbb{R}^{N_2}$. For a concrete chamber Γ , let $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^N$ be a linear isomorphism given in Theorem 2.2 so that $\Phi(\Gamma) = \Omega$.

It suffices to estimate an operator with integral kernel

$$f(\Phi^{-1}x)K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space $L_2(\Omega, w(\Phi^{-1}x)dx)$. Using Lemma 3.5, it suffices to estimate an operator with integral kernel

$$f(\Phi^{-1}x)K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y), \quad x, y \in \Omega,$$

on the Hilbert space $L_2(\Omega, w_\theta(x)dx)$.

By Lemma 5.3,

$$\left| \partial_x^\alpha \partial_y^\beta K_{\text{Dunkl}^{-\frac{1}{2}}}(\Phi^{-1}x, g\Phi^{-1}y) \right| \leq c_{\alpha, \beta, \kappa} \frac{|x - y|^{1 - |\alpha|_1 - |\beta|_1}}{w_\theta(B(x, |x - y|))}, \quad x, y \in \Omega.$$

The other assumptions in Theorem 4.1 follow from Lemma 5.4, Lemma 3.3 and Lemma 3.4. The assertion follows now from Theorem 4.1. $\square$

6. PROOF OF THEOREM 1.1 (i)

Lemma 6.1. *We have*

$$[T_{e_j}, M_f] = M_{\partial_j f} + \frac{1}{2} \sum_{v \in R} v_j \kappa(v) M_{\delta_v f} O_v,$$

where

$$(\delta_v f)(x) = \frac{f(x) - (O_v f)(x)}{\langle v, x \rangle}, \quad v \in R, \quad x \in \mathbb{R}^N,$$

and where O_v is (the operation of composition with) reflection with respect to vector v .

Proof. Obvious. $\square$

Lemma 6.2. *We have*

$$\|\delta_v f\|_{L_N(\mathbb{R}^N)} \leq c_N \|f\|_{\dot{W}^{1, N}(\mathbb{R}^N)}.$$

Proof. Without loss of generality, $v = e_N$. Set

$$(D_t h)(x) = h(x_1, \dots, x_{N-1}, tx_N).$$

We have

$$\delta_v f = \int_0^1 D_t(\partial_N f) dt.$$

Thus,

$$\begin{aligned} \|\delta_v f\|_{L_N(\mathbb{R}^N)} &\leq \int_0^1 \|D_t(\partial_N f)\|_{L_N(\mathbb{R}^N)} dt = \\ &= \int_0^1 t^{-\frac{1}{N}} \|\partial_N f\|_{L_N(\mathbb{R}^N)} dt = \frac{N}{N-1} \|\partial_N f\|_{L_N(\mathbb{R}^N)}. \end{aligned}$$

$\square$

Lemma 6.3. *For every $N \geq 1$, we have*

$$\|[B, A]B^{-1}\|_{N, \infty} \leq c_N \|B^{-1}[B^2, A]B^{-1}\|_{N, \infty}.$$

Proof of Theorem 1.1 (i). We prove the assertion under *a priori* restriction $f \in C_c^\infty(\mathbb{R}^N)$. Passing to the case of $f \in \dot{W}^{1,N}(\mathbb{R}^N)$ is standard (see e.g. [refer to our papers](#)).

We have

$$\begin{aligned} [R_{\text{Dunkl},j}, M_f] &= [T_{e_j} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}, M_f] = [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} + T_{e_j} [\Delta_{\text{Dunkl}}^{-\frac{1}{2}}, M_f] = \\ &= [T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} - R_{\text{Dunkl},j} [\Delta_{\text{Dunkl}}^{\frac{1}{2}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}. \end{aligned}$$

By triangle inequality,

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|[T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} \|[\Delta_{\text{Dunkl}}^{\frac{1}{2}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Using Lemma 6.3, we write

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|[T_{e_j}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} c_N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Note that

$$\begin{aligned} \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}} &= \\ &= \sum_{k=1}^N \Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f] R_{\text{Dunkl},k} + R_{\text{Dunkl},k} [T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}. \end{aligned}$$

By triangle inequality,

$$\begin{aligned} \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [\Delta_{\text{Dunkl}}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \sum_{k=1}^N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ \frac{N}{N-1} \sum_{k=1}^N \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Combining the estimates in the last two paragraphs, we obtain

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq c'_N \sum_{k=1}^N \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} [T_{e_k}, M_f]\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ &+ c'_N \sum_{k=1}^N \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Using Lemma 6.1, the fact that Δ_{Dunkl} commutes with O_v and the triangle inequality, we obtain

$$\begin{aligned} \|[T_{e_k}, M_f] \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} &\leq \\ &\leq \frac{N}{N-1} \|M_{\partial_k} f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \end{aligned}$$

$$+ \frac{N}{2(N-1)} \sum_{v \in R} |v| \kappa(v) \|M_{\delta_v} f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}.$$

Thus,

$$\begin{aligned} & \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ & \leq c_N'' \sum_{k=1}^N \|M_{\partial_k} f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} + \\ & + c_\kappa \sum_{v \in R} \|M_{\delta_v} f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

The assertion follows now from Theorem 5.5 and Lemma 6.2. $\square$

7. PROOF OF THEOREM 1.1 (ii)

Lemma 7.1. *Let $1 \leq j \leq N$. If $x_j \geq y_j$, then*

$$K_{\text{Dunkl},j}(x, y) \geq c_{R,k} \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

If $x_j \leq y_j$, then

$$K_{\text{Dunkl},j}(x, y) \leq c_{R,k} \frac{x_j - y_j}{|x - y|} \cdot \frac{1}{w(B(x, |x - y|))}.$$

Proof. The proof is identical to that in [6, Theorem 1.2]. $\square$

Lemma 7.2. *Let $Q \in \mathbb{D}$. We have*

$$K_j(x, y) \geq \frac{c_k}{w(Q)}, \quad x \in 3Q, \quad y \in Q + 3s(Q)e_j.$$

Proof. The assertion follows immediately from Lemma 7.1. $\square$

Lemma 7.3. *Let real-valued $f \in L_1^{\text{loc}}(\mathbb{R}^N)$ be such that the commutator $[R_{\text{Dunkl},j}, M_f]$ is bounded on $L_2(\mathbb{R}^N, w(x)dx)$. For every $Q \in \mathbb{D}$, there exist ξ_Q and η_Q supported in $7Q$ such that $\|\xi_Q\|_\infty, \|\eta_Q\|_\infty \leq m(Q)^{-\frac{1}{2}}$ and such that*

$$\frac{1}{w(3Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq c_k |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

Proof. Let α be the median of f on $3Q$ and β be the median of f on $Q + 3s(Q)e_j$. Assume for definiteness $\alpha \geq \beta$. Set

$$\begin{aligned} A_1 &= \{x \in 3Q : f(x) \geq \alpha\}, \quad B_1 = \{x \in Q + 3s(Q)e_j : f(x) \leq \beta\}, \\ A_2 &= \{x \in 3Q : f(x) \leq \beta\}, \quad B_2 = \{x \in Q + 3s(Q)e_j : f(x) \geq \beta\}, \\ \xi_{Q,s} &= w(Q)^{-\frac{1}{2}} \chi_{A_s}, \quad \eta_{Q,s} = w(Q)^{-\frac{1}{2}} \chi_{B_s}, \quad s = 1, 2. \end{aligned}$$

We have

$$\begin{aligned} & \langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} = \\ & = \frac{1}{w(Q)} \int_{B_s \times A_s} K_j(x, y) (f(y) - f(x)) w(x) w(y) dx dy. \end{aligned}$$

If $x \in B_1$ and $y \in A_1$, then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \alpha) + (\alpha - \beta) + (\beta - f(x)) = \\ &= |f(y) - \alpha| + |\alpha - \beta| + |f(x) - \beta| \geq |f(y) - \alpha| + |\alpha - \beta| \geq 0. \end{aligned}$$

If $x \in B_2$ and $y \in A_2$, then

$$\begin{aligned} f(y) - f(x) &= (f(y) - \beta) + (\beta - f(x)) = \\ &= -|f(y) - \beta| - |f(x) - \beta| \leq -|f(y) - \beta| \leq 0. \end{aligned}$$

Thus,

$$\begin{aligned} &\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ &\geq \frac{1}{w(Q)} \int_{B_1 \times A_1} K_j(x, y) (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy \geq \\ &\geq \frac{c_k}{w^2(Q)} \int_{B_1 \times A_1} (|f(y) - \alpha| + |\alpha - \beta|) w(x) w(y) dx dy = \\ &= \frac{c_k}{w^2(Q)} \cdot \left(w(B_1) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(B_1) w(A_1) \right) \geq \\ &= \frac{c_k}{w^2(Q)} \cdot \left(\frac{1}{2} w(Q) \cdot \int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| \cdot \frac{1}{4} w^2(Q) \right) \geq \\ &\geq \frac{c_k}{4w(Q)} \cdot \left(\int_{A_1} |f(y) - \alpha| w(y) dy + |\alpha - \beta| w(Q) \right) \end{aligned}$$

and

$$\begin{aligned} &-\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \geq \\ &\geq \frac{1}{w(Q)} \int_{B_2 \times A_2} K_j(x, y) |f(y) - \beta| w(x) w(y) dx dy \geq \\ &\geq \frac{c_k}{w^2(Q)} \int_{B_2 \times A_2} |f(y) - \beta| w(x) w(y) dx dy = \\ &= \frac{c_k}{w^2(Q)} \cdot w(B_2) \cdot \int_{A_2} |f(y) - \beta| w(y) dy \geq \\ &= \frac{c_k}{2w(Q)} \int_{A_2} |f(y) - \beta| w(y) dy. \end{aligned}$$

It remains to note that

$$\begin{aligned} &\int_{3Q} |f(y) - \alpha| w(y) dy = \int_{A_1} |f(y) - \alpha| w(y) dy + \\ &+ \int_{A_2} |f(y) - \alpha| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |f(y) - \alpha| w(y) dy \leq \\ &\leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy \\ &+ \int_{A_2} |\alpha - \beta| w(y) dy + \int_{3Q \setminus (A_1 \cup A_2)} |\alpha - \beta| w(y) dy \leq \\ &\leq \int_{A_1} |f(y) - \alpha| w(y) dy + \int_{A_2} |f(y) - \beta| w(y) dy + 2|\alpha - \beta| w(3Q). \end{aligned}$$

Combining the last 3 equations, we conclude

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq \max_{s=1,2} |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,s}, \eta_{Q,s} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

If

$$|\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,1}, \eta_{Q,1} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}| \geq |\langle [R_{\text{Dunkl},j}, M_f] \xi_{Q,2}, \eta_{Q,2} \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|,$$

then we set $\xi_Q = \xi_{Q,1}$ and $\eta = \eta_{Q,1}$. Otherwise, set $\xi_Q = \xi_{Q,2}$ and $\eta_Q = \eta_{Q,2}$. Thus,

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - \alpha| w(y) dy \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

Hence,

$$c'_k |f_{3Q} - \alpha| \leq |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|$$

and, finally,

$$\frac{c'_k}{w(Q)} \int_{3Q} |f(y) - f_{3Q}| w(y) dy \leq 2 |\langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)}|.$$

□

Lemma 7.4. *Suppose that $f \in L_1^{\text{loc}}(\mathbb{R}^N)$. If $[R_{\text{Dunkl},j}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$ for some $j \in \{1, 2, \dots, N\}$, then*

$$\begin{aligned} & \left\| \left\{ \frac{1}{w(3Q)} \int_{3Q} |f(x) - f_{3Q}| w(x) dx \right\}_{Q \in \mathbb{D}} \right\|_{l_{N,\infty}(\mathbb{D})} \leq \\ & \leq c_k \| [R_{\text{Dunkl},j}, M_f] \|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}. \end{aligned}$$

Proof. Let $\{\xi_Q\}_{Q \in \mathbb{D}}$ and $\{\eta_Q\}_{Q \in \mathbb{D}}$ be functions constructed in Lemma 7.3. By Lemma 2.6,

$$\left\{ \langle [R_{\text{Dunkl},j}, M_f] \xi_Q, \eta_Q \rangle_{L_2(\mathbb{R}^N, w(x)dx)} \right\}_{Q \in \mathbb{D}} \in l_p.$$

The first assertion follows now from Lemma 7.3. □

Lemma 7.5. *Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). We have*

$$\sup_{x \in Q} w(x) \leq \frac{c_k}{m(Q)} \int_Q w(x) dx.$$

Proof. If $x = c_Q + y$, $y \in [-\frac{1}{2}s(Q), \frac{1}{2}s(Q)]^d$, then

$$|\langle x, v \rangle| \leq |\langle c_Q, v \rangle| + r |\langle y, v \rangle| \leq |\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q).$$

Thus,

$$w(x) = \prod_{v \in R} |\langle x, v \rangle|^{k(v)} \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q))^{k(v)}.$$

In particular,

$$LHS \leq \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{\sqrt{d}}{2} s(Q))^{k(v)}.$$

Fix chamber Γ such that $c_Q \in \Gamma$. We have

$$\int_Q w(x) dx \geq \int_{c_Q + (B(0, \frac{1}{2}s(Q)) \cap \Gamma)} w(x) dx = \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} w(c_Q + y) dy.$$

If $y \in \Gamma$, then

$$|\langle c_Q + y, v \rangle| = |\langle c_Q, v \rangle| + |\langle y, v \rangle|, \quad v \in R.$$

Thus,

$$w(c_Q + y) = \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)}.$$

Hence,

$$\int_Q w(x)dx \geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma} \prod_{v \in R} (|\langle c_Q, v \rangle| + |\langle y, v \rangle|)^{k(v)} dy.$$

Fix a closed cone $\Gamma' \subset \Gamma$ such that (i) interior of Γ' is non-empty and (ii) $\partial\Gamma \cap \partial\Gamma' = \{0\}$. We have

$$|\langle y, v \rangle| \geq c_{\Gamma'} |y|, \quad y \in \Gamma', \quad v \in R.$$

We, therefore, have

$$\begin{aligned} \int_Q w(x)dx &\geq \int_{B(0, \frac{1}{2}s(Q)) \cap \Gamma'} \prod_{v \in R} (|\langle c_Q, v \rangle| + c_{\Gamma'} |y|)^{k(v)} dy \geq \\ &\geq \int_{\substack{y \in \Gamma' \\ \frac{1}{4}s(Q) \leq |y| \leq \frac{1}{4}s(Q)}} \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{2}c_{\Gamma'} s(Q))^{k(v)} dy = \\ &= s(Q)^N \prod_{v \in R} (|\langle c_Q, v \rangle| + \frac{1}{4}c_{\Gamma'} s(Q))^{k(v)} \times (2^{-N} - 4^{-N}) \int_{\substack{y \in \Gamma' \\ |y| \leq 1}} dy. \end{aligned}$$

Combining this with the preceding paragraph, we complete the proof. $\square$

Lemma 7.6. *Suppose $1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} k(v)}$. Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). We have*

$$\left(\frac{\int_Q w^{1-t}(x)dx}{\int_Q w(x)dx} \right)^{\frac{1}{t}} \leq c_{t,k} \frac{m(Q)}{\int_Q w(x)dx}.$$

Proof. If $v \in R$ is such that

$$\sqrt{d}s(Q) \leq |\langle c_Q, v \rangle|,$$

then

$$|\langle x, v \rangle| \geq |\langle c_Q, v \rangle| - |\langle x - c_Q, v \rangle| \geq |\langle c_Q, v \rangle| - \frac{\sqrt{d}}{2}s(Q) \geq \frac{1}{2}|\langle c_Q, v \rangle|, \quad x \in Q.$$

In this case,

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx &\leq 2^{(t-1)|R|k(v)} |\langle c_Q, v \rangle|^{(1-t)|R|k(v)} \cdot \int_Q dx = \\ &= 2^{(t-1)|R|k(v)} |\langle x_0, v \rangle|^{(1-t)|R|k(v)} \cdot s(Q)^N. \end{aligned}$$

If $v \in R$ is such that

$$|\langle c_Q, v \rangle| \leq \sqrt{d}s(Q),$$

then let $y_0 = \langle c_Q, v \rangle$ and $z_0 = c_Q - y_0 v$. Let us introduce new variables $y = \langle x, v \rangle$ and $z = x - yv$ (it varies in the plane orthogonal to v). We write

$$\begin{aligned} \int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx &\leq \int_{|x - c_Q| \leq \frac{\sqrt{d}}{2}s(Q)} |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \leq \\ &\leq \int_{|y - y_0|, |z - z_0| \leq \frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy dz = \\ &= \left(\frac{\sqrt{d}}{2}s(Q) \right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y - y_0| \leq \frac{\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy \leq \\ &= \left(\frac{\sqrt{d}}{2}s(Q) \right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}s(Q)} |y|^{(1-t)|R|k(v)} dy = \end{aligned}$$

$$= s(Q)^{N+(1-t)|R|k(v)} \cdot \left(\frac{\sqrt{d}}{2}\right)^{N-1} \text{Vol}(\mathbb{B}^{N-1}) \cdot \int_{|y| \leq \frac{3\sqrt{d}}{2}} |y|^{(1-t)|R|k(v)} dy.$$

Combining the estimates in the preceding paragraphs, we write

$$\int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \leq c_1 s(Q)^N (\max\{|\langle c_Q, v \rangle|, r\})^{(1-t)|R|k(v)}.$$

By Hölder inequality,

$$\begin{aligned} \int_Q w^{1-t}(x) dx &\leq \prod_{v \in R} \left(\int_Q |\langle x, v \rangle|^{(1-t)|R|k(v)} dx \right)^{\frac{1}{|R|}} \leq \\ &\leq c_2 s(Q)^N \prod_{v \in R} (\max\{|\langle c_Q, v \rangle|, s(Q)\})^{(1-t)k(v)}. \end{aligned}$$

The assertion follows easily. $\square$

Lemma 7.7. *Measures dx and $w(x)dx$ (and distance $(x, y) \rightarrow |x - y|_\infty$) satisfy the conditions in Proposition 1.2 in [9].*

Proof. That both measures are doubling is obvious.

Let Q be a cube in $\mathbb{R}^N$ (with sides parallel to coordinate hyperplanes). By Lemmas 7.5 and 7.6,

$$\begin{aligned} \left(\frac{1}{m(Q)} \int_Q w^t(x) dx \right)^{\frac{1}{t}} &\leq \frac{c_k}{m(Q)} \int_Q w(x) dx, \quad t > 0. \\ \left(\frac{\int_Q w^{1-t}(x) dx}{\int_Q w(x) dx} \right)^{\frac{1}{t}} &\leq c_G \frac{m(Q)}{\int_Q w(x) dx}, \quad 1 < t < 1 + \frac{1}{|R| \cdot \max_{v \in R} k(v)}. \end{aligned}$$

The assertion follows now from Proposition 3.6 in [9]. $\square$

Proof of Theorem 1.1 (ii). The left hand side in Lemma 7.4 is the oscillation semi-norm of the function f with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$. So, the oscillation semi-norm of the function f with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$ is controlled by the $\|\cdot\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator $[R_{\text{Dunkl},j}, M_f]$.

Lemma 7.7 asserts that measures dx and $w(x)dx$ (and distance $(x, y) \rightarrow |x - y|_\infty$) satisfy the conditions in Proposition 1.2 in [9]. Appealing to Proposition 1.2 in [9], we conclude that the oscillation semi-norm of the function f with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$ is controlled by the one with respect to measure $w(x)dx$ and distance $(x, y) \rightarrow |x - y|_\infty$. By the preceding paragraph, the oscillation semi-norm of the function f with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$ is controlled by the $\|\cdot\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))}$ -norm of the commutator $[R_{\text{Dunkl},j}, M_f]$.

That oscillation semi-norm of the function f with respect to measure dx and distance $(x, y) \rightarrow |x - y|_\infty$ is equivalent to the Sobolev semi-norm of the function f is a standard fact. The proof (in much more general setting) is available in Proposition 1.29 in [10]. The assertion follows now from the preceding paragraph. $\square$

8. PROOF OF SMOOTHNESS

Theorem 8.1. *Let $f \in C_c^\infty(\mathbb{R}^N)$.*

(i) *for every $n \in \mathbb{N}$, $s_1, s_2 \in \mathbb{R}$ with $s_1 + s_2 = n$, the operator¹*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

is bounded on $L_2(\mathbb{R}^N, w(x)dx)$;

(ii) *for every $n \in \mathbb{N}$, $s_1, s_2 \in \mathbb{R}$ with $s_1 + s_2 > n$, the operator²*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{s_1+s_2-n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$;

Lemma 8.2. *We have*

$$\begin{aligned} [\Delta_{\text{Dunkl}}, M_f] &= \sum_{j=1}^N M_{f_j^{(1)}} T_{e_j} + \sum_{j=1}^N \sum_{v \in R} c_{j,v} M_{f_v^{(1)}} (T_{e_j} O_v + O_v T_{e_j}) + \\ &\quad + M_{f^{(2)}} + \sum_{v \in R} M_{f_v^{(2)}} O_v + \sum_{v, v' \in R} M_{f_{v,v'}} O_{v'} O_v. \end{aligned}$$

All the functions on the right hand side are smooth and compactly supported.

Proof. We have

$$\begin{aligned} [\Delta_{\text{Dunkl}}, M_f] &= \sum_{j=1}^N T_{e_j} [T_{e_j}, M_f] + [T_{e_j}, M_f] T_{e_j} = \\ &= \sum_{j=1}^N T_{e_j} M_{\partial_j f} + M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (T_{e_j} M_{\delta_v f} O_v + M_{\delta_v f} O_v T_{e_j}) = \\ &= 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (M_{\delta_v f} T_{e_j} O_v + M_{\delta_v f} O_v T_{e_j}) + \\ &\quad + \sum_{j=1}^N [T_{e_j}, M_{\partial_j f}] + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) [T_{e_j}, M_{\delta_v f}] O_v = \\ &= 2 \sum_{j=1}^N M_{\partial_j f} T_{e_j} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) (M_{\delta_v f} T_{e_j} O_v + M_{\delta_v f} O_v T_{e_j}) + \\ &\quad + \sum_{j=1}^N M_{\partial_j^2 f} + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} \alpha_j \kappa(v) M_{\delta_v(\partial_j f)} O_v + \\ &\quad + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) M_{\partial_j(\delta_v f)} O_v + \frac{1}{4} \sum_{j=1}^N \sum_{v, v' \in R} v_j v'_j \kappa(v) \kappa(v') M_{\delta_{v'} \delta_v f} O_{v'} O_v. \end{aligned}$$

□

¹Commutators are repeated n times

²Commutators are repeated n times

Lemma 8.3. *For every $f \in C_c^\infty(\mathbb{R}^N)$, the operator*

$$[\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$$

is bounded and

$$(1 + \Delta_{\text{Dunkl}})^{-1}[\Delta_{\text{Dunkl}}, M_f] \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. We prove only the first assertion. The proof of the second one is similar (modulo Theorem 5.5).

Since each T_{e_j} commutes with $(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$, each O_v commutes with $(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$ and since the operators

$$\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, \quad 1 \leq j \leq N,$$

are bounded, the assertion follows from Lemma 8.2. $\square$

Lemma 8.4. *For every $n \in \mathbb{N}$, the operator³*

$$[\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]](1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}}$$

is bounded.

Proof. Denote for brevity

$$\text{Expr}_n(f) = [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]](1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}}.$$

We prove the assertion by induction on f . The n -th statement is "for every $f \in C_c^\infty(\mathbb{R}^N)$, the operator $\text{Expr}_n(f)$ is bounded." Base of induction is given by Lemma 8.3. It suffices to prove the step of induction.

Since each T_{e_j} and each O_v commutes with Δ_{Dunkl} , it follows from Lemma 8.2 that

$$\begin{aligned} \text{Expr}_{n+1}(f) &= \\ &= \sum_{j=1}^N \text{Expr}_n(f_j^{(1)}) \cdot \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \\ &+ \sum_{j=1}^N \sum_{v \in R} c_{j,v} \text{Expr}_n(f_v^{(1)}) \cdot \left(\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} O_v + O_v \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \right) + \\ &+ \text{Expr}_n(f^{(2)}) \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \sum_{v \in R} \text{Expr}_n(f_v^{(2)}) \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} O_v + \\ &+ \sum_{v,v' \in R} \text{Expr}_n(f_{v,v'}^{(2)}) \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} O_{v'} O_v. \end{aligned}$$

In each summand, the first factor is bounded by inductive assumption and the second factor is obviously bounded. This yields the step of induction and, hence, completes the proof. $\square$

Proof of Theorem 8.1 (i). The assertion follows from Lemma 8.4 and Hadamard 3 lines theorem. $\square$

Lemma 8.5. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-1} \in \mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

³Commutators are repeated n times

Proof. Set

$$\mathcal{D} = \sum_{j=1}^N \gamma_j \otimes T_{e_j},$$

where $\{\gamma_j\}_{j=1}^N$ are Pauli matrices. We have

$$\begin{aligned} 1 \otimes M_f(1 + \Delta_{\text{Dunkl}})^{-1} &= (1 \otimes M_f)(\mathcal{D}^2 + 1)^{-1} = (1 \otimes M_f)(\mathcal{D} + i)^{-1}(\mathcal{D} - i)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + [1 \otimes M_f, (\mathcal{D} + i)^{-1}](\mathcal{D} - i)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + (\mathcal{D} + i)^{-1}[\mathcal{D}, 1 \otimes M_f](\mathcal{D}^2 + 1)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + (\mathcal{D} + i)^{-1}\left(\sum_{j=1}^N \gamma_j \otimes [T_{e_j}, M_f]\right)(\mathcal{D}^2 + 1)^{-1} = \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + \sum_{j=1}^N (\mathcal{D} + i)^{-1}(\gamma_j \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1} + \\ &\quad + \sum_{j=1}^N \sum_{\alpha \in R_+} \alpha_j \kappa(\alpha)(\mathcal{D} + i)^{-1}(\gamma_j \otimes M_{f_\alpha} O_\alpha)(\mathcal{D}^2 + 1)^{-1}. \end{aligned}$$

Since Δ_{Dunkl} commutes with O_α , it follows that

$$\begin{aligned} 1 \otimes M_f(1 + \Delta_{\text{Dunkl}})^{-1} &= \\ &= (\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1} + \sum_{j=1}^N (\mathcal{D} + i)^{-1}(1 \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1} \cdot (\gamma_j \otimes 1) + \\ &\quad + \sum_{j=1}^N \sum_{\alpha \in R_+} \alpha_j \kappa(\alpha)(\mathcal{D} + i)^{-1}(1 \otimes M_{f_\alpha})(\mathcal{D}^2 + 1)^{-1} \cdot (\gamma_j \otimes O_\alpha). \end{aligned}$$

Clearly,

$$\begin{aligned} &\|(\mathcal{D} + i)^{-1}(1 \otimes M_f)(\mathcal{D} - i)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} = \\ &= \|(\Delta_{\text{Dunkl}} + 1)^{-\frac{1}{2}} M_f (\Delta_{\text{Dunkl}} + 1)^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ &= \|\Delta_{\text{Dunkl}}^{-\frac{1}{2}} M_f \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ &\leq 2^{\frac{2}{N}} \|M_{|f|^{\frac{1}{2}}} \Delta_{\text{Dunkl}}^{-\frac{1}{2}}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))}^2 \leq \\ &\stackrel{\text{Th. 5.5}}{\leq} c_{N, \kappa} \| |f|^{\frac{1}{2}} \|_{L_N(\mathbb{R}^N)}^2 = c_{N, \kappa} \|f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)} \end{aligned}$$

and, similarly,

$$\begin{aligned} &\|(\mathcal{D} + i)^{-1}(1 \otimes M_{\partial_j f})(\mathcal{D}^2 + 1)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} \leq c_{N, \kappa} \|\partial_j f\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}, \\ &\|(\mathcal{D} + i)^{-1}(1 \otimes M_{f_\alpha})(\mathcal{D}^2 + 1)^{-1}\|_{\mathcal{L}_{\frac{N}{2}, \infty}(\mathbb{C}^{2^{\lfloor \frac{N}{2} \rfloor}} \otimes L_2(\mathbb{R}^N, w(x)dx))} \leq c_{N, \kappa} \|f_\alpha\|_{L_{\frac{N}{2}}(\mathbb{R}^N)}. \end{aligned}$$

This completes the proof. $\square$

Lemma 8.6. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \in \mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. We prove the assertion by induction on f . The n -th statement is "for every $f \in C_c^\infty(\mathbb{R}^N)$, we have

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n}{2}} \in \mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))."$$

Base of induction (i.e. cases $n = 1$ and $n = 2$) are given by Theorem 5.5 and Lemma 8.5. It suffices to prove the step of induction.

We have

$$\begin{aligned} & M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-1} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + [M_f, (1 + \Delta_{\text{Dunkl}})^{-1}](1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-1} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + (1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}}. \end{aligned}$$

Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $(f \circ U_g) \cdot \phi = f \circ U_g$ for every $g \in G$. It follows that

$$[\Delta_{\text{Dunkl}}, M_f] = [\Delta_{\text{Dunkl}}, M_f] M_\phi.$$

Thus,

$$\begin{aligned} & M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{n+1}{2}} = (1 + \Delta_{\text{Dunkl}})^{-1} M_f \cdot M_\phi(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} + \\ &+ (1 + \Delta_{\text{Dunkl}})^{-1} [\Delta_{\text{Dunkl}}, M_f] \cdot M_\phi(1 + \Delta_{\text{Dunkl}})^{-\frac{n-1}{2}} \cdot (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}. \end{aligned}$$

In the first summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{\frac{N}{2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.5 and the second one belongs to the ideal $\mathcal{L}_{\frac{N}{n-1}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption. In the second summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.3, the second factor belongs to the ideal $\mathcal{L}_{\frac{N}{n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption and the third one is obviously bounded. $\square$

Lemma 8.7. *Let $k, l \in \mathbb{Z}_+$ with $l > k$. For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{l-k}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$.

Proof. We prove the assertion by induction on k . Base of induction (i.e., the case $k = 0$) is established in Lemma 8.6.

We have

$$\begin{aligned} & T_{e_{m_1}} \cdots T_{e_{m_k}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} = T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_f T_{e_{m_k}}(1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} + \\ &+ T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} [T_{e_{m_k}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}} = \\ &= T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times \frac{T_{e_{m_k}}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} + \\ &+ T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_{\partial_{m_k} f}(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} + \\ &+ \frac{1}{2} \sum_{v \in R} v_{m_k} \kappa(v) T_{e_{m_1}} \cdots T_{e_{m_{k-1}}} M_{\delta_v f}(1 + \Delta_{\text{Dunkl}})^{-\frac{l-1}{2}} \times (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} O_v. \end{aligned}$$

In each summand on the right hand side, the first factor belongs to the ideal $\mathcal{L}_{\frac{N}{l-k}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by inductive assumption and the second factor is obviously bounded. This yields the step of induction and, hence, completes the proof. $\square$

Lemma 8.8. *Let $s_1, s_2 \in \mathbb{R}$ be such that $s_1 + s_2 > 0$. We have*

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}$$

belongs to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$.

Proof. Step 1: Let us prove the assertion for $s_1, s_2 \in \mathbb{Z}$.

Suppose first $s_1 \leq 0$. Denote $k = -s_1 \geq 0$ and $l = s_2 > k$. Denote for brevity

$$V = (1 + \Delta_{\text{Dunkl}})^{\frac{k}{2}} (1 + \sum_{j_1, \dots, j_k=1}^N |T_{e_{j_1}} \cdots T_{e_{j_k}}|)^{-1}.$$

Obviously, V is bounded. We have

$$\begin{aligned} (1 + \Delta_{\text{Dunkl}})^{\frac{k}{2}} &= V \times (1 + \sum_{m_1, \dots, m_k=1}^N |T_{e_{m_1}} \cdots T_{e_{m_k}}|) = V + \\ &+ \sum_{m_1, \dots, m_k=1}^N V \times \text{sgn}(T_{e_{m_1}} \cdots T_{e_{m_k}})^* \times T_{e_{m_1}} \cdots T_{e_{m_k}}. \end{aligned}$$

It, therefore, suffices to prove that operators

$$M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}, \quad T_{e_{m_1}} \cdots T_{e_{m_k}} M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{l}{2}}$$

belong to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. This is established in Lemma 8.6 and Lemma 8.7, respectively. Hence, the claim is proved for $s_1, s_2 \in \mathbb{Z}$ and $s_1 \leq 0$.

If $s_2 \leq 0$, then the claim follows from that for $s_2 \leq 0$ by taking the adjoint. If $s_1, s_2 \geq 0$, then, according to Lemma 8.7 the operators

$$(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1+s_2}{2}} M_f, \quad M_f (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1+s_2}{2}}$$

belong to $\mathcal{L}_{\frac{N}{s_1+s_2}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$. The claim for $s_1, s_2 \geq 0$ follows now from the Hadamard 3 lines theorem.

Step 2: The general case follows now from Step 1 by complex interpolation. $\square$

Proof of Theorem 8.1 (ii). Choose $\phi \in C_c^\infty(\mathbb{R}^N)$ such that $(f \circ U_g) \cdot \phi = f \circ U_g$ for every $g \in G$. It follows that

$$[\Delta_{\text{Dunkl}}, M_f] = [\Delta_{\text{Dunkl}}, M_f] M_\phi.$$

We have

$$\begin{aligned} &(1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{s}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] M_\phi (1 + \Delta_{\text{Dunkl}})^{-\frac{s}{2}} = \\ &= (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1}{2}} [\Delta_{\text{Dunkl}}, [\Delta_{\text{Dunkl}}, [\cdots, M_f] \cdots]] (1 + \Delta_{\text{Dunkl}})^{-\frac{n-s_1}{2}} \times \\ &\quad \times (1 + \Delta_{\text{Dunkl}})^{-\frac{s_1-n}{2}} M_\phi (1 + \Delta_{\text{Dunkl}})^{-\frac{s_2}{2}}. \end{aligned}$$

The first factor is bounded by Theorem 8.1 (i). The second factor belongs to $\mathcal{L}_{\frac{N}{s_1+s_2-n}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.8. $\square$

9. FIRST APPROXIMATION THEOREM

Theorem 9.1. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$\begin{aligned} & [(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}, M_f](1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \frac{1}{2}[\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} \in \\ & \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx)). \end{aligned}$$

Lemma 9.2. *Let $B \geq 1$. Suppose*

$$B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

We have

$$[B, A]B^{-1} - \frac{1}{2}[B^2, A]B^{-2} \in \mathcal{L}_{\frac{2N}{3}, \infty}.$$

Proof. We have

$$B = \frac{2}{\pi} \int_0^\infty \frac{B^2 d\lambda}{\lambda^2 + B^2}.$$

Hence,

$$[B, A]B^{-1} = \frac{2}{\pi} \int_0^\infty \left[\frac{B^2}{\lambda^2 + B^2}, A \right] B^{-1} d\lambda.$$

Obviously,

$$\begin{aligned} \left[\frac{B^2}{\lambda^2 + B^2}, A \right] B^{-1} &= - \left[\frac{\lambda^2}{\lambda^2 + B^2}, A \right] B^{-1} = \frac{\lambda^2}{\lambda^2 + B^2} [B^2, A] \frac{1}{B(\lambda^2 + B^2)} = \\ &= [B^2, A] \frac{\lambda^2}{B(\lambda^2 + B^2)^2} - \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2}. \end{aligned}$$

Obviously,

$$\begin{aligned} \int_0^\infty [B^2, A] \frac{\lambda^2}{B(\lambda^2 + B^2)^2} d\lambda &= [B^2, A] B^{-1} \cdot \int_0^\infty \frac{\lambda^2}{(\lambda^2 + B^2)^2} d\lambda = \\ &= [B^2, A] B^{-2} \cdot \int_0^\infty \frac{\lambda^2}{(\lambda^2 + 1)^2} d\lambda = \frac{\pi}{4} [B^2, A] B^{-2}. \end{aligned}$$

On the other hand,

$$\begin{aligned} & \left\| \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} \right\|_{\frac{2N}{3}, \infty} \leq \\ & \leq \left\| \frac{\lambda^2 B}{\lambda^2 + B^2} \right\|_\infty \|B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}}\|_{\frac{2N}{3}, \infty} \left\| \frac{B^{\frac{3}{2}}}{(\lambda^2 + B^2)^2} \right\|_\infty. \end{aligned}$$

Next,

$$\left\| \frac{\lambda^2 B}{\lambda^2 + B^2} \right\|_\infty \leq \sup_{t \geq 1} \frac{\lambda^2 t}{\lambda^2 + t^2} \leq \lambda$$

and

$$\left\| \frac{B^{\frac{3}{2}}}{(\lambda^2 + B^2)^2} \right\|_\infty \leq \sup_{t \geq 1} \frac{t^{\frac{3}{2}}}{(\lambda^2 + t^2)^2} \leq (\lambda^2 + 1)^{-\frac{5}{4}}.$$

Thus,

$$\begin{aligned} & \left\| \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} \right\|_{\frac{2N}{3}, \infty} \leq \\ & \leq \frac{\lambda}{(\lambda^2 + 1)^{\frac{5}{4}}} \|B^{-1}[B^2, [B^2, A]]B^{-\frac{5}{2}}\|_{\frac{2N}{3}, \infty}. \end{aligned}$$

Hence, the integral

$$\int_0^\infty \frac{\lambda^2}{\lambda^2 + B^2} [B^2, [B^2, A]] \frac{1}{B(\lambda^2 + B^2)^2} d\lambda$$

converges absolutely in $\mathcal{L}_{\frac{2N}{3}, \infty}$ and the assertion follows. $\square$

Proof of Theorem 9.1. The assertion follows from Theorem 8.1 and Lemma 9.2. $\square$

10. SECOND APPROXIMATION THEOREM

Theorem 10.1. *If $f \in C_c^\infty(\mathbb{R}^N)$, then*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in (\mathcal{L}_{N, \infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Lemma 10.2. *If P is a differential operator of degree 1 with smooth compactly supported coefficients and if the coefficients are supported away from the hyperplanes $\{x : \langle x, v \rangle = 0\}$, $v \in R$, then*

$$\int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} d\lambda \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Fix $\psi \in C_c^\infty(\mathbb{R}^N)$ such that $P = PM_\psi$. We have $P = M_{w^{-\frac{1}{2}}} P' M_{w^{\frac{1}{2}}}$, where P' is a differential operator of degree 1 with smooth compactly supported coefficients. We write

$$\begin{aligned} & \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} = \\ & = \left(M_{w^{-\frac{1}{2}}} \frac{1}{1 + \lambda^2 + \Delta} P' M_{w^{\frac{1}{2}}} \right) \times M_\psi \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}}. \end{aligned}$$

Hence,

$$\begin{aligned} & \left\| \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} \right\|_{\mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ & \leq 2^{\frac{3}{2N}} \left\| M_{w^{-\frac{1}{2}}} \frac{1}{1 + \lambda^2 + \Delta} P' M_{w^{\frac{1}{2}}} \right\|_{\mathcal{L}_{2N, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \times \\ & \quad \times \left\| M_\psi \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} \right\|_{\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ & \leq 2^{\frac{3}{2N}} \left\| \frac{1}{1 + \lambda^2 + \Delta} P' \right\|_{\mathcal{L}_{2N, \infty}(L_2(\mathbb{R}^N))} \times \\ & \quad \times (1 + \lambda^2)^{-\frac{1}{2}} \left\| M_\psi \Delta_{\text{Dunkl}}^{-\frac{1}{2}} \right\|_{\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq \\ & \leq \frac{c_{P'}}{(1 + \lambda^2)^{\frac{1}{4}}} \times \frac{c_\psi}{(1 + \lambda^2)^{\frac{1}{2}}}. \end{aligned}$$

Hence, the integral converges absolutely. $\square$

Lemma 10.3. *If P is a differential operator of degree 1 with smooth compactly supported coefficients and if the coefficients are supported away from the hyperplanes $\{x : \langle x, v \rangle = 0\}$, $v \in R$, then*

$$\int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} d\lambda \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. We have $P = M_{w^{-\frac{1}{2}}} P' M_{w^{\frac{1}{2}}}$, where P' is a differential operator of degree 1 with smooth compactly supported coefficients. Therefore,

$$\begin{aligned} & \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} = \\ & = M_{w^{-\frac{1}{2}}} \times \frac{1}{1 + \lambda^2 + \Delta} P' \frac{1}{1 + \lambda^2 + \Delta} \times M_{w^{\frac{1}{2}}}. \end{aligned}$$

Hence,

$$\begin{aligned} & \left\| \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \right\|_{\mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx))} = \\ & = \left\| \frac{1}{1 + \lambda^2 + \Delta} P' \frac{1}{1 + \lambda^2 + \Delta} \right\|_{\mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N))} \leq \frac{c_{P'}}{(1 + \lambda^2)^{\frac{3}{4}}}. \end{aligned}$$

Hence, our integral converges absolutely. $\square$

Lemma 10.4. *If $f \in C_c^\infty(\mathbb{R}^N)$ is supported away from the hyperplanes $\{x : \langle x, v \rangle = 0\}$, $v \in R$, then*

$$M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. We have

$$\begin{aligned} & M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} = \\ & = \frac{2}{\pi} \int_0^\infty M_f \left(\frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} - \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \right) d\lambda. \end{aligned}$$

Obviously,

$$\begin{aligned} & M_f \left(\frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} - \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \right) = \\ & = M_f \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \left(M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}} - \Delta_{\text{Dunkl}} \right) \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} = \\ & = \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} M_f \left(M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}} - \Delta_{\text{Dunkl}} \right) \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} + \\ & + \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} [M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f] \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \times \\ & \times \left(M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}} - \Delta_{\text{Dunkl}} \right) \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} = \\ & = \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} M_f \left(M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}} - \Delta_{\text{Dunkl}} \right) \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} + \\ & + \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} [M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f] \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} - \\ & - \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} [M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f] \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}}. \end{aligned}$$

We have

$$\begin{aligned} & M_f \left(\Delta_{\text{Dunkl}} - M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}} \right) + [M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f] = \\ & = P + \sum_{v \in R} P_v O_v + \sum_{v, v' \in R} P_{v, v'} O_v O_{v'}, \end{aligned}$$

where P , $\{P_v\}_{v \in R}$ and $\{P_{v,v'}\}_{v,v' \in R}$ are differential operators of degree 1 with smooth compactly supported coefficients. Commutator $[M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}, M_f] = P'$, where P' is a differential operator of degree 1 with smooth compactly supported coefficients. Moreover, coefficients of P , P' and $\{P_v\}_{v \in R}$ are supported away from the hyperplanes $\{x : \langle x, v \rangle = 0\}_{v \in R}$. Since Δ_{Dunkl} commutes with O_v , $v \in R$, it follows that

$$\begin{aligned} M_f \left(\frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} - \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \right) &= \\ &= \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} + \\ &+ \sum_{v \in R} \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P_v \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} \times O_v + \\ &+ \sum_{v,v' \in R} \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P_{v,v'} \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} \times O_v O_{v'} - \\ &- \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P' \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}}. \end{aligned}$$

Thus,

$$\begin{aligned} M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} &= \\ &= \frac{2}{\pi} \int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} d\lambda + \\ &+ \frac{2}{\pi} \sum_{v \in R} \int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P_v \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} d\lambda \times O_v + \\ &+ \frac{2}{\pi} \sum_{v,v' \in R} \int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P_{v,v'} \frac{1}{1 + \lambda^2 + \Delta_{\text{Dunkl}}} d\lambda \times O_v O_{v'} - \\ &- \frac{2}{\pi} \int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P' \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} d\lambda. \end{aligned}$$

The assertion follows now from Lemmas 10.2 and 10.3. $\square$

Proof of Theorem 10.1. Choose a sequence $\{\phi_n\}_{n \geq 1} \subset C_c^\infty(\mathbb{R}^N)$ such that (a) $0 \leq \phi_n \leq 1$ for $n \geq 1$ (b) ϕ_n is supported away from the hyperplanes $\{x : \langle x, v \rangle = 0\}$, $v \in R$, for $n \geq 1$ (c) $\text{supp}(\phi_n) \uparrow \mathbb{R}^N$ as $n \rightarrow \infty$.

By Lemma 10.4,

$$M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in \mathcal{L}_{\frac{2N}{3}, \infty}(L_2(\mathbb{R}^N, w(x)dx))$$

for every $n \geq 1$.

By Theorem 5.5,

$$M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} \rightarrow M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}$$

in $\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ as $n \rightarrow \infty$. By the usual Cwikel estimate,

$$M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \rightarrow M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}$$

in $\mathcal{L}_{N, \infty}(L_2(\mathbb{R}^N, w(x)dx))$ as $n \rightarrow \infty$.

Thus,

$$M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} \in (\mathcal{L}_{N, \infty})_0(L_2(\mathbb{R}^N, w(x)dx))$$

for every $n \geq 1$ and

$$\begin{aligned} \lim_{n \rightarrow \infty} M_{f\phi_n}(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_{f\phi_n}(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} = \\ = M_f(1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - M_f(1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}}, \end{aligned}$$

where the limit is understood in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$. Since the separable part $(\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$ is closed in $\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$, the assertion follows. $\square$

11. FINAL APPROXIMATION THEOREM

Theorem 11.1. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$[R_{\text{Dunkl},j}, M_f] - \textcolor{red}{insert} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Lemma 11.2. *We have*

$$[R_{\text{Dunkl},j}, M_f] - [\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f] \in \mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. Set

$$h(x) = 1 - \frac{|x|}{(1 + |x|^2)^{\frac{1}{2}}}, \quad x \in \mathbb{R}^N.$$

We, obviously, have

$$R_{\text{Dunkl},j} - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} = R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}).$$

Thus,

$$\begin{aligned} [R_{\text{Dunkl},j}, M_f] - [\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f] = \\ = R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}) M_f - M_f h(\nabla_{\text{Dunkl}}) \cdot R_{\text{Dunkl},j} = \\ = R_{\text{Dunkl},j} \cdot h(\nabla_{\text{Dunkl}}) (1 + \Delta_{\text{Dunkl}}) \cdot (1 + \Delta_{\text{Dunkl}})^{-1} M_f - \\ - M_f (1 + \Delta_{\text{Dunkl}})^{-1} \cdot (1 + \Delta_{\text{Dunkl}}) h(\nabla_{\text{Dunkl}}) \cdot R_{\text{Dunkl},j}. \end{aligned}$$

In the first summand on the right hand side, the first and the second factors are bounded and the third one belongs to $\mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.6. In the second summand on the right hand side, the second and the third factors are bounded and the first one belongs to $\mathcal{L}_{\frac{N}{2},\infty}(L_2(\mathbb{R}^N, w(x)dx))$ by Lemma 8.6. $\square$

Lemma 11.3. *For every $f \in C_c^\infty(\mathbb{R}^N)$, we have*

$$[\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} - \textcolor{red}{insert} \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)).$$

Proof. We have

$$\begin{aligned} & [\Delta_{\text{Dunkl}}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} = \\ & = \sum_{j=1}^N T_{e_j} [T_{e_j}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} + [T_{e_j}, M_f] T_{e_j} (1 + \Delta_{\text{Dunkl}})^{-1} = \\ & = \sum_{j=1}^N T_{e_j} [T_{e_j}, M_f](1 + \Delta_{\text{Dunkl}})^{-1} + [T_{e_j}, M_f] \frac{\Delta_{\text{Dunkl}}^{\frac{1}{2}}}{1 + \Delta_{\text{Dunkl}}} \times R_{\text{Dunkl},j} = \\ & = \sum_{j=1}^N T_{e_j} M_{\partial_j f} (1 + \Delta_{\text{Dunkl}})^{-1} + M_{\partial_j f} \frac{\Delta_{\text{Dunkl}}^{\frac{1}{2}}}{1 + \Delta_{\text{Dunkl}}} \times R_{\text{Dunkl},j} + \end{aligned}$$

$$\begin{aligned}
& + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) \times T_{e_j} M_{\delta_v f} (1 + \Delta_{\text{Dunkl}})^{-1} \times O_v + \\
& + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) \times M_{\delta_v f} \frac{\Delta_{\text{Dunkl}}^{\frac{1}{2}}}{1 + \Delta_{\text{Dunkl}}} \times O_v \times R_{\text{Dunkl},j} = \\
& = \sum_{j=1}^N [T_{e_j}, M_{\partial_j f}] (1 + \Delta_{\text{Dunkl}})^{-1} + 2M_{\partial_j f} \frac{\Delta_{\text{Dunkl}}^{\frac{1}{2}}}{1 + \Delta_{\text{Dunkl}}} \times R_{\text{Dunkl},j} + \\
& + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) \times [T_{e_j}, M_{\delta_v f}] (1 + \Delta_{\text{Dunkl}})^{-1} \times O_v + \\
& + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) \times M_{\delta_v f} T_{e_j} (1 + \Delta_{\text{Dunkl}})^{-1} \times O_v + \\
& + \frac{1}{2} \sum_{j=1}^N \sum_{v \in R} v_j \kappa(v) \times M_{\delta_v f} \frac{\Delta_{\text{Dunkl}}^{\frac{1}{2}}}{1 + \Delta_{\text{Dunkl}}} \times O_v \times R_{\text{Dunkl},j} =
\end{aligned}$$

□

Proof of Theorem 11.1. We have

$$\begin{aligned}
& \left[\frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}}, M_f \right] = \\
& = [T_{e_j}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot [(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}}.
\end{aligned}$$

Using Lemma 11.2 and Theorem 9.1, we write

$$\begin{aligned}
& [R_{\text{Dunkl},j}, M_f] \in [T_{e_j}, M_f] (1 + \Delta_{\text{Dunkl}})^{-\frac{1}{2}} - \\
& - \frac{1}{2} \cdot \frac{T_{e_j}}{(1 + \Delta_{\text{Dunkl}})^{\frac{1}{2}}} \cdot [\Delta_{\text{Dunkl}}, M_f] (1 + \Delta_{\text{Dunkl}})^{-1} \in (\mathcal{L}_{N,\infty})_0 (L_2(\mathbb{R}^N, w(x)dx)).
\end{aligned}$$

□

12. SPECTRAL ASYMPTOTIC FORMULA

Definition 12.1. For $g \in G$ and $j = 1, \dots, N$, set

$$d_{g,j} := g^{-1} e_j, \quad f_g(x) := f(gx).$$

For $\alpha \in R_+$, define $S_\alpha : \mathbb{C}^G \rightarrow \mathbb{C}^G$ by

$$(S_\alpha z)_g := z_{g\sigma_\alpha}, \quad g \in G.$$

For $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, $x \in \Gamma_+$, and $\theta \in \mathbb{S}^{N-1}$, define

$$\begin{aligned}
\mathfrak{c}_{f,j}(x, \theta) &:= \text{diag}_{g \in G} \left(\langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right) \\
&+ \sum_{\alpha \in R_+} k(\alpha) \text{diag}_{g \in G} \left(\frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha.
\end{aligned}$$

For smooth f , the quotients in the second term are understood by their smooth extensions across the corresponding reflection hyperplanes. For $f \in \dot{W}^{1,N}(\mathbb{R}^N)$, $\mathfrak{c}_{f,j}$ is understood almost everywhere on $\Gamma \times \mathbb{S}^{N-1}$.

Lemma 12.2. *Let Γ_+ be an open Weyl chamber, let Γ be its closure, and let R_+ be the corresponding positive subsystem. Set*

$$\rho(x) := \varpi(x)^{1/2},$$

and define

$$\begin{aligned} U : L^2(\mathbb{R}^N, dw) &\longrightarrow L^2(\Gamma, dw; \mathbb{C}^G), & (Uu)_g(x) &:= u(gx), \\ V : L^2(\Gamma, dw; \mathbb{C}^G) &\longrightarrow L^2(\Gamma, dx; \mathbb{C}^G), & (Vh)(x) &:= \rho(x)h(x). \end{aligned}$$

Then U and V are unitary. If

$$\tilde{T}_j := VUT_{e_j}U^{-1}V^{-1}, \quad \tilde{P} := VU(-\Delta_{\text{Dunkl}})U^{-1}V^{-1},$$

then, on $C_c^\infty(\Gamma_+; \mathbb{C}^G)$,

$$(1) \quad (\tilde{T}_j h)_g = \partial_{d_{g,j}} h_g - \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} h_{g\sigma_\alpha},$$

$$(2) \quad \tilde{P} = -\Delta_{\mathbb{C}^G} + |\nabla \log \rho|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2}.$$

Assume in addition that $N > 1$, and let

$$\tilde{Q} := VU(-\Delta_{\text{Dunkl}})^{-1/2}U^{-1}V^{-1},$$

where $(-\Delta_{\text{Dunkl}})^{-1/2}$ is defined by the spectral calculus. Then, for every $\chi \in C_c^\infty(\Gamma_+)$, the operator $M_\chi \tilde{Q} M_\chi$, initially defined on its natural domain, admits a bounded extension to $L^2(\Gamma, dx; \mathbb{C}^G)$, and this extension is a classical matrix-valued pseudodifferential operator of order -1 . Its principal symbol is

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = \chi(x)^2 |\xi|^{-1} I_{\mathbb{C}^G}.$$

Consequently, if $x \in \Gamma_+$, $\xi \neq 0$, and $\chi \in C_c^\infty(\Gamma_+)$ is identically 1 in a neighborhood of x , then

$$(3) \quad \sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G}.$$

Proof. The Weyl chambers $g\Gamma_+$, $g \in G$, are pairwise disjoint and cover $\mathbb{R}^N$ up to the reflection hyperplanes; see [7, Theorem 2.12]. Since dw is G -invariant,

$$\begin{aligned} \|Uu\|_{L^2(\Gamma, dw; \mathbb{C}^G)}^2 &= \sum_{g \in G} \int_{\Gamma} |u(gx)|^2 dw(x) \\ &= \int_{\mathbb{R}^N} |u(x)|^2 dw(x). \end{aligned}$$

Conversely, for $h = (h_g)_{g \in G} \in L^2(\Gamma, dw; \mathbb{C}^G)$, define $u(gx) := h_g(x)$ for $x \in \Gamma_+$. This determines u almost everywhere on $\mathbb{R}^N$ and gives $Uu = h$. Hence U is unitary. Since $dw = \rho^2 dx$ and $\rho > 0$ on Γ_+ , V is unitary as well.

Pairing the roots α and $-\alpha$ in the definition of T_{e_j} , changing variables in the root system by g , and using the G -invariance of k , we obtain, for $h \in C_c^\infty(\Gamma_+; \mathbb{C}^G)$,

$$(UT_{e_j}U^{-1}h)_g = \partial_{d_{g,j}} h_g + \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (h_g - h_{g\sigma_\alpha}).$$

On Γ_+ ,

$$\nabla \log \rho(x) = \sum_{\alpha \in R_+} k(\alpha) \frac{\alpha}{\langle \alpha, x \rangle},$$

and therefore

$$\rho \partial_{d_{g,j}} \rho^{-1} = \partial_{d_{g,j}} - \sum_{\alpha \in R_+} k(\alpha) \frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle}.$$

Conjugation by V cancels the diagonal terms and gives (1).

Similarly, pairing α and $-\alpha$ in the formula for the Dunkl Laplacian and using $\|\alpha\|^2 = 2$, we have

$$\begin{aligned} (U(-\Delta_{\text{Dunkl}})U^{-1}h)_g &= -\Delta h_g - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{\partial_\alpha h_g}{\langle \alpha, x \rangle} \\ &\quad + 2 \sum_{\alpha \in R_+} k(\alpha) \frac{h_g - h_{g\sigma_\alpha}}{\langle \alpha, x \rangle^2}. \end{aligned}$$

Writing $\phi = \log \rho$,

$$\rho(-\Delta - 2\nabla\phi \cdot \nabla)\rho^{-1} = -\Delta + |\nabla\phi|^2 + \Delta\phi,$$

while

$$\Delta\phi = - \sum_{\alpha \in R_+} k(\alpha) \frac{\|\alpha\|^2}{\langle \alpha, x \rangle^2} = -2 \sum_{\alpha \in R_+} \frac{k(\alpha)}{\langle \alpha, x \rangle^2}.$$

The latter term cancels the diagonal part of

$$2 \sum_{\alpha \in R_+} k(\alpha) \frac{I_{\mathbb{C}^G} - S_\alpha}{\langle \alpha, x \rangle^2},$$

and hence

$$\tilde{P} = -\Delta I_{\mathbb{C}^G} + |\nabla \log \rho|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2},$$

which proves (2).

Assume now that $N > 1$, fix $\chi \in C_c^\infty(\Gamma_+)$, and set

$$K := \text{supp } \chi.$$

On Γ_+ , write

$$\tilde{P} = -\Delta I_{\mathbb{C}^G} + W(x), \quad W(x) = |\nabla \log \rho(x)|^2 I_{\mathbb{C}^G} - 2 \sum_{\alpha \in R_+} k(\alpha) \frac{S_\alpha}{\langle \alpha, x \rangle^2}.$$

Thus W is a smooth Hermitian matrix-valued function on Γ_+ .

Choose open sets

$$K \Subset \Omega_0 \Subset \Omega_1 \Subset \Gamma_+$$

and $\eta \in C_c^\infty(\Gamma_+)$ such that

$$0 \leq \eta \leq 1, \quad \eta = 1 \quad \text{on a neighborhood of } \overline{\Omega_1}.$$

Choose a flat torus M containing $\text{supp } \eta$ in one Euclidean coordinate chart. Since η is compactly supported in that chart, the differential operator $M_\eta \tilde{P} M_\eta$ extends smoothly by zero to M . Define on $L^2(M; \mathbb{C}^G)$

$$(4) \quad P_0 := M_\eta \tilde{P} M_\eta + M_{1-\eta}(-\Delta_M + 1)M_{1-\eta}.$$

It is a nonnegative self-adjoint elliptic differential operator of order 2. Indeed,

$$\langle P_0 u, u \rangle = \langle \tilde{P}(\eta u), \eta u \rangle + \|\nabla((1-\eta)u)\|_2^2 + \|(1-\eta)u\|_2^2 \geq 0,$$

and its principal symbol is

$$p_{0,2}(x, \xi) = (\eta(x)^2 + (1-\eta(x))^2)|\xi|^2 I_{\mathbb{C}^G},$$

which is elliptic since

$$\eta^2 + (1-\eta)^2 \geq \frac{1}{2}.$$

Moreover,

$$(5) \quad P_0 = \tilde{P} \quad \text{on } \Omega_1.$$

Let

$$H_t := e^{-t\tilde{P}}, \quad H_t^0 := e^{-tP_0}.$$

We first consider

$$B_0 := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} H_t^0 dt.$$

Let Π_0 be the orthogonal projection onto $\ker P_0$. Since P_0 is elliptic on the closed manifold M , Π_0 is smoothing. Applying the complex powers calculus [?, Theorem 2 and p. 290] to $P_0 + \Pi_0$, we obtain

$$(P_0 + \Pi_0)^{-1/2} \in \Psi_{\text{cl}}^{-1}(M; \mathbb{C}^G)$$

with principal symbol $p_{0,2}^{-1/2}$.

Set

$$F(\lambda) := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} e^{-t\lambda} dt, \quad \lambda \geq 0.$$

Then $B_0 = F(P_0)$, and for $\lambda > 0$,

$$F(\lambda) - \lambda^{-1/2} = -\frac{1}{\sqrt{\pi}} \int_1^\infty t^{-1/2} e^{-t\lambda} dt.$$

Hence

$$F(\lambda) - \lambda^{-1/2} = O(\lambda^{-m}) \quad (\lambda \rightarrow \infty)$$

for every $m > 0$. On $\ker P_0$ the difference between B_0 and $(P_0 + \Pi_0)^{-1/2}$ is finite rank. It follows from elliptic spectral regularity that

$$B_0 - (P_0 + \Pi_0)^{-1/2} \in \Psi^{-\infty}(M; \mathbb{C}^G).$$

Consequently,

$$(6) \quad B_0 \in \Psi_{\text{cl}}^{-1}(M; \mathbb{C}^G), \quad \sigma_{-1}(B_0)(x, \xi) = p_{0,2}(x, \xi)^{-1/2}.$$

In particular, by (5),

$$(7) \quad \sigma_{-1}(B_0)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G}, \quad x \in \Omega_1.$$

We next compare H_t and H_t^0 in the interior. Choose $\zeta \in C_c^\infty(\Omega_1)$ such that

$$\zeta = 1 \quad \text{on a neighborhood of } K.$$

Since $P_0 = \tilde{P}$ on Ω_1 , Duhamel's formula gives

$$(8) \quad M_\chi(H_t - H_t^0)M_\chi = - \int_0^t M_\chi H_{t-s} [\tilde{P}, M_\zeta] H_s^0 M_\chi ds.$$

Indeed, for data supported in K ,

$$v(t) := M_\zeta H_t^0 M_\chi u$$

satisfies

$$(\partial_t + \tilde{P})v = [\tilde{P}, M_\zeta] H_t^0 M_\chi u, \quad v(0) = M_\chi u,$$

which yields (8).

Let $h_t(x, y)$ be the Dunkl heat kernel. The integral kernel of H_t , with respect to Lebesgue measure on Γ , is the matrix

$$(9) \quad \tilde{h}_t(x, y)_{g, \ell} = \rho(x)\rho(y) h_t(gx, \ell y), \quad g, \ell \in G.$$

For $x, y \in \Gamma$,

$$d(gx, \ell y) = d(x, y) = |x - y|.$$

Together with the mixed derivative estimate for the Dunkl heat kernel [2, Theorem 3.1], this implies that, whenever $E, F \Subset \Gamma_+$ and $\text{dist}(E, F) > 0$, for every pair of multi-indices α, β and every $m > 0$,

$$(10) \quad \sup_{\substack{x \in E \\ y \in F}} \left\| \partial_x^\alpha \partial_y^\beta \tilde{h}_t(x, y) \right\| = O(t^m), \quad t \downarrow 0.$$

The heat kernel of the smooth elliptic operator P_0 satisfies the same off-diagonal estimate on M .

Since

$$\text{supp}[\tilde{P}, M_\zeta] \cap K = \emptyset,$$

the two heat kernels occurring in (8) are evaluated between compact sets separated by a positive distance. The standard Gaussian bounds underlying (10) therefore give, for every multi-indices α, β and every $m > 0$,

$$(11) \quad \sup_{x, y} \left\| \partial_x^\alpha \partial_y^\beta K_{M_\chi(H_t - H_t^0)M_\chi}(x, y) \right\| = O(t^m), \quad t \downarrow 0.$$

Hence

$$(12) \quad S_0 := \frac{1}{\sqrt{\pi}} \int_0^1 t^{-1/2} M_\chi(H_t - H_t^0)M_\chi dt \in \Psi^{-\infty}.$$

It remains to treat the large-time part. Let $K_1 \Subset \Gamma_+$ be a compact set containing K . By (9), the heat-kernel derivative estimate cited above, and (2), for $x, y \in K_1$ and $t \geq 1$,

$$V(gx, \ell y, \sqrt{t})^{-1} \lesssim_{K_1} t^{-N/2}.$$

Since ρ and all its derivatives are bounded on K_1 , it follows that, for every pair of multi-indices α, β ,

$$(13) \quad \sup_{x, y \in K_1} \left\| \partial_x^\alpha \partial_y^\beta \tilde{h}_t(x, y) \right\| \lesssim_{\alpha, \beta, K_1} t^{-N/2}, \quad t \geq 1.$$

Since $N > 1$,

$$\int_1^\infty t^{-1/2} t^{-N/2} dt < \infty.$$

Thus all derivatives of

$$K_{S_\infty}(x, y) := \frac{\chi(x)\chi(y)}{\sqrt{\pi}} \int_1^\infty t^{-1/2} \tilde{h}_t(x, y) dt$$

converge locally uniformly, and therefore

$$(14) \quad S_\infty := \frac{1}{\sqrt{\pi}} \int_1^\infty t^{-1/2} M_\chi H_t M_\chi dt \in \Psi^{-\infty}.$$

For completeness, the preceding integrals give the canonical bounded localization of $\tilde{P}^{-1/2}$. Indeed, for $\varepsilon > 0$,

$$(\tilde{P} + \varepsilon)^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} e^{-\varepsilon t} H_t dt$$

by the spectral theorem. The estimates (11) and (13) imply that

$$M_\chi (\tilde{P} + \varepsilon)^{-1/2} M_\chi$$

converges in operator norm, as $\varepsilon \downarrow 0$, to

$$\frac{1}{\sqrt{\pi}} \int_0^\infty t^{-1/2} M_\chi H_t M_\chi dt.$$

On the natural domain of $\tilde{P}^{-1/2}$, this limit agrees with $M_\chi \tilde{P}^{-1/2} M_\chi$; hence it is its bounded extension.

Combining (12), (14), and the definition of B_0 , we obtain

$$M_\chi \tilde{Q} M_\chi = M_\chi B_0 M_\chi + S_0 + S_\infty.$$

By (6), $M_\chi B_0 M_\chi$ is a classical matrix-valued pseudodifferential operator of order -1 , while $S_0 + S_\infty$ is smoothing. Therefore

$$M_\chi \tilde{Q} M_\chi \in \Psi_{\text{cl}}^{-1},$$

and, by (7),

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = \chi(x)^2 |\xi|^{-1} I_{\mathbb{C}^G}.$$

If $\chi = 1$ in a neighborhood of x , this reduces to

$$\sigma_{-1}(M_\chi \tilde{Q} M_\chi)(x, \xi) = |\xi|^{-1} I_{\mathbb{C}^G},$$

which proves (3). □

Lemma 12.3. *Let $j \in \{1, \dots, N\}$ and $f \in C_c^\infty(\mathbb{R}^N)$, and set*

$$A_{f,j} := VU[R_{\text{Dunkl},j}, M_f]U^{-1}V^{-1}.$$

Then, for every $\chi \in C_c^\infty(\Gamma_+)$,

$$M_\chi A_{f,j} M_\chi$$

is a compactly supported classical matrix valued pseudodifferential operator of order -1 , with principal symbol

$$(15) \quad \sigma_{-1}(M_\chi A_{f,j} M_\chi)(x, \xi) = \frac{\chi(x)^2}{\|\xi\|} \mathbf{c}_{f,j}\left(x, \frac{\xi}{\|\xi\|}\right), \quad x \in \Gamma_+, \quad \xi \neq 0.$$

Proof. Set

$$F(x) := \text{diag}_{g \in G} (f_g(x)).$$

Since $VUM_fU^{-1}V^{-1} = M_F$ and $R_{\text{Dunkl},j} = T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$,

$$(16) \quad A_{f,j} = [\tilde{T}_j, M_F] \tilde{Q} + \tilde{T}_j [\tilde{Q}, M_F].$$

By (1),

$$(17) \quad \begin{aligned} [\tilde{T}_j, M_F] &= \text{diag}_{g \in G} (\langle d_{g,j}, \nabla f_g \rangle) \\ &+ \sum_{\alpha \in R_+} k(\alpha) \text{diag}_{g \in G} \left(\frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g - f_{g\sigma_\alpha}) \right) S_\alpha. \end{aligned}$$

Fix $\chi \in C_c^\infty(\Gamma_+)$. Choose $\eta, \psi \in C_c^\infty(\Gamma_+)$ such that $\eta = 1$ on a neighborhood of $\text{supp } \chi$ and $\psi = 1$ on a neighborhood of $\text{supp } \eta$, and set

$$Q_\psi := M_\psi \tilde{Q} M_\psi.$$

By Lemma 12.2, Q_ψ is a classical matrix valued pseudodifferential operator of order -1 , and on a neighborhood of $\text{supp } \eta$,

$$(18) \quad q_{-1}(x, \xi) = \|\xi\|^{-1} I_{\mathbb{C}^G}.$$

We use the Kohn–Nirenberg convention

$$\text{Op}(a)u(x) = (2\pi)^{-N} \int_{\mathbb{R}^{2N}} e^{i(x-y) \cdot \xi} a(x, \xi) u(y) dy d\xi.$$

If a and b are classical symbols, the composition formula [?, Theorem 18.1.8, p. 71] gives

$$(19) \quad a \# b \sim \sum_{\beta \in \mathbb{N}_0^N} \frac{1}{\beta!} \partial_\xi^\beta a D_x^\beta b, \quad D_x := \frac{1}{i} \partial_x.$$

Write, on a neighborhood of $\text{supp } \eta$,

$$\sigma(Q_\psi)(x, \xi) \sim q_{-1}(x, \xi) + q_{-2}(x, \xi) + \cdots.$$

By (2),

$$\sigma(\tilde{P})(x, \xi) \sim p_2(x, \xi) + p_1(x, \xi) + p_0(x, \xi),$$

where

$$p_2(x, \xi) = \|\xi\|^2 I_{\mathbb{C}^G}, \quad p_1(x, \xi) = 0.$$

Since $\tilde{Q} = \tilde{P}^{-1/2}$, $\tilde{Q}\tilde{P}\tilde{Q} = I$. Applying (19) on the region where $\psi = 1$ and comparing the homogeneous terms of degree -1 , the terms containing x -derivatives of p_2 and q_{-1} vanish, and hence

$$q_{-2}p_2q_{-1} + q_{-1}p_2q_{-2} = 0.$$

Together with (18), this gives

$$(20) \quad q_{-2}(x, \xi) = 0$$

on a neighborhood of $\text{supp } \eta$.

Applying (19) to the commutator $[Q_\psi, M_F]$, and using (18)–(20), we obtain, on the same neighborhood,

$$\begin{aligned} \sigma_{-2}([Q_\psi, M_F])(x, \xi) &= \frac{1}{i} \sum_{\ell=1}^N \partial_{\xi_\ell} (\|\xi\|^{-1}) \partial_{x_\ell} F(x) \\ (21) \qquad \qquad \qquad &= i \|\xi\|^{-3} \operatorname{diag}_{g \in G} (\langle \xi, \nabla f_g(x) \rangle). \end{aligned}$$

Since $\psi = 1$ on a neighborhood of $\operatorname{supp} \eta$ and $\eta = 1$ on a neighborhood of $\operatorname{supp} \chi$,

$$\begin{aligned} M_\chi[\tilde{T}_j, M_F] \tilde{Q} M_\chi &= M_\chi[\tilde{T}_j, M_F] Q_\psi M_\chi, \\ M_\chi \tilde{T}_j[\tilde{Q}, M_F] M_\chi &= M_\chi \tilde{T}_j M_\eta [Q_\psi, M_F] M_\chi. \end{aligned}$$

Both operators on the right are classical matrix valued pseudodifferential operators of order -1 .

By (17), (18), and the order-zero part of the composition formula,

$$\begin{aligned} \sigma_{-1}(M_\chi[\tilde{T}_j, M_F] \tilde{Q} M_\chi)(x, \xi) &= \frac{\chi(x)^2}{\|\xi\|} \operatorname{diag}_{g \in G} (\langle d_{g,j}, \nabla f_g(x) \rangle) \\ (22) \qquad \qquad \qquad &+ \frac{\chi(x)^2}{\|\xi\|} \sum_{\alpha \in R_+} k(\alpha) \operatorname{diag}_{g \in G} \left(\frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha. \end{aligned}$$

Under the above Kohn–Nirenberg convention, (1) gives

$$\sigma_1(\tilde{T}_j)(x, \xi) = i \operatorname{diag}_{g \in G} (\langle d_{g,j}, \xi \rangle).$$

Therefore, by (21),

$$\begin{aligned} \sigma_{-1}(M_\chi \tilde{T}_j[\tilde{Q}, M_F] M_\chi)(x, \xi) \\ (23) \qquad \qquad \qquad &= -\chi(x)^2 \|\xi\|^{-3} \operatorname{diag}_{g \in G} (\langle d_{g,j}, \xi \rangle \langle \xi, \nabla f_g(x) \rangle). \end{aligned}$$

Combining (22) and (23), and setting

$$\theta := \frac{\xi}{\|\xi\|},$$

we obtain

$$\begin{aligned} \sigma_{-1}(M_\chi A_{f,j} M_\chi)(x, \xi) &= \frac{\chi(x)^2}{\|\xi\|} \operatorname{diag}_{g \in G} \left(\langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right) \\ &+ \frac{\chi(x)^2}{\|\xi\|} \sum_{\alpha \in R_+} k(\alpha) \operatorname{diag}_{g \in G} \left(\frac{\langle \alpha, d_{g,j} \rangle}{\langle \alpha, x \rangle} (f_g(x) - f_{g\sigma_\alpha}(x)) \right) S_\alpha \\ &= \frac{\chi(x)^2}{\|\xi\|} \mathbf{c}_{f,j}(x, \theta). \end{aligned}$$

This proves (15). $\square$

Lemma 12.4. *Assume that $N > 1$, $j \in \{1, \dots, N\}$, and $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. Then*

$$\left(\int_\Gamma \int_{\mathbb{S}^{N-1}} \operatorname{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \simeq_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

Proof. By the definition of $\mathbf{c}_{f,j}$, the unitary invariance of the Schatten norm, and $\|d_{g,j}\| = 1$,

$$\begin{aligned} \|\mathbf{c}_{f,j}(x, \theta)\|_{S^N} &\lesssim_{R,k} \left(\sum_{g \in G} |\nabla f_g(x)|^N \right)^{1/N} \\ &\quad + \sum_{\alpha \in R_+} \left(\sum_{g \in G} \left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right|^N \right)^{1/N}. \end{aligned}$$

The chamber decomposition used in Lemma 12.2 and the orthogonality of $g \in G$ imply

$$\sum_{g \in G} \int_{\Gamma} |\nabla f_g(x)|^N dx = \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}^N.$$

Moreover, since $\|\alpha\|^2 = 2$,

$$|x - \sigma_\alpha x| = \sqrt{2} |\langle \alpha, x \rangle|,$$

and hence

$$\left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right| = \sqrt{2} \mathcal{G}_{g, g\sigma_\alpha} f(x)$$

for almost every $x \in \mathbb{R}^N$. By Lemma ??,

$$\int_{\Gamma} \left| \frac{f_g(x) - f_{g\sigma_\alpha}(x)}{\langle \alpha, x \rangle} \right|^N dx \lesssim_{N,G} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}^N$$

uniformly in $g \in G$ and $\alpha \in R_+$. Since G and R_+ are finite,

$$(24) \quad \left(\int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \lesssim_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

For the converse estimate, let P_g denote the orthogonal projection of $\mathbb{C}^G$ onto the g -th coordinate and define

$$\mathcal{E}(A) := \sum_{g \in G} P_g A P_g, \quad A \in \mathcal{L}(\mathbb{C}^G).$$

Thus, $\mathcal{E}(A)$ is the diagonal part of A . The pinching inequality for unitarily invariant norms [?, p. 97, (IV.51)–(IV.52)] gives

$$(25) \quad \|\mathcal{E}(A)\|_{S^N} \leq \|A\|_{S^N}.$$

For every $\alpha \in R_+$, S_α has zero diagonal, since $g\sigma_\alpha \neq g$ for all $g \in G$. Hence, by the definition of $\mathbf{c}_{f,j}$,

$$\mathcal{E}(\mathbf{c}_{f,j}(x, \theta)) = \text{diag}_{g \in G} \left(\langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right).$$

Therefore, by (25),

$$(26) \quad \|\mathbf{c}_{f,j}(x, \theta)\|_{S^N}^N \geq \sum_{g \in G} \left| \langle d_{g,j}, \nabla f_g(x) \rangle - \langle d_{g,j}, \theta \rangle \langle \theta, \nabla f_g(x) \rangle \right|^N.$$

For each $g \in G$, set

$$y := gx, \quad \eta := g\theta.$$

Since

$$\nabla f_g(x) = g^{-1} \nabla f(y), \quad d_{g,j} = g^{-1} e_j,$$

and g is orthogonal,

$$\begin{aligned} \langle d_{g,j}, \nabla f_g(x) \rangle &= \partial_j f(y), \\ \langle d_{g,j}, \theta \rangle &= \eta_j, \\ \langle \theta, \nabla f_g(x) \rangle &= \langle \eta, \nabla f(y) \rangle. \end{aligned}$$

Using the chamber decomposition and the orthogonal invariance of surface measure, (26) therefore gives

$$\begin{aligned} (27) \quad & \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \|\mathbf{c}_{f,j}(x, \theta)\|_{S^N}^N d\theta dx \\ & \geq \int_{\mathbb{R}^N} \int_{\mathbb{S}^{N-1}} |\partial_j f(y) - \eta_j \langle \eta, \nabla f(y) \rangle|^N d\eta dy. \end{aligned}$$

For $v \in \mathbb{C}^N$, set

$$\Phi_j(v) := \left(\int_{\mathbb{S}^{N-1}} |v_j - \eta_j \langle \eta, v \rangle|^N d\eta \right)^{1/N}.$$

Then

$$(28) \quad \Phi_j(v) \simeq_N \|v\|.$$

Indeed, Φ_j is a seminorm on $\mathbb{C}^N$. If $\Phi_j(v) = 0$, continuity gives

$$v_j - \eta_j \langle \eta, v \rangle = 0, \quad \eta \in \mathbb{S}^{N-1}.$$

Since $N > 1$, there exists $\eta \in \mathbb{S}^{N-1}$ with $\eta_j = 0$, and therefore $v_j = 0$. It follows that

$$\langle \eta, v \rangle = 0$$

whenever $\eta_j \neq 0$. Since such η form a dense subset of $\mathbb{S}^{N-1}$, continuity yields $\langle \eta, v \rangle = 0$ for every $\eta \in \mathbb{S}^{N-1}$, and hence $v = 0$. Thus Φ_j is a norm, and (28) follows from the equivalence of norms on the finite dimensional space $\mathbb{C}^N$.

Applying (28) with $v = \nabla f(y)$ in (27), we obtain

$$\|\nabla f\|_{L^N(\mathbb{R}^N, dx)} \lesssim_N \left(\int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Combining this estimate with (24) proves the assertion. $\square$

Lemma 12.5. *Assume that $N > 1$, $j \in \{1, \dots, N\}$, and $f \in C_c^\infty(\mathbb{R}^N)$. Then*

$$(29) \quad \lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_f]) = \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Proof. Define

$$A_f := VU[R_{\text{Dunkl},j}, M_f]U^{-1}V^{-1}.$$

By Lemma ?? (??), there exist $\chi_m \in C_c^\infty(\Gamma_+)$, $0 \leq \chi_m \leq 1$, such that $\chi_m(x) \rightarrow 1$ for every $x \in \Gamma_+$ and

$$(30) \quad \|A_f - M_{\chi_m} A_f M_{\chi_m}\|_{S^{N,\infty}} \rightarrow 0.$$

Set

$$A_{f,m} := M_{\chi_m} A_f M_{\chi_m}, \quad K_m := \text{supp } \chi_m.$$

By Lemma 12.3 (15), $A_{f,m}$ is a compactly supported classical matrix valued pseudodifferential operator of order -1 on Γ_+ , with

$$(31) \quad \sigma_{-1}(A_{f,m})(x, \xi) = \frac{\chi_m(x)^2}{\|\xi\|} \mathbf{c}_{f,j} \left(x, \frac{\xi}{\|\xi\|} \right), \quad x \in \Gamma_+, \quad \xi \neq 0.$$

Choose a bounded open set U_m with $K_m \Subset U_m \Subset \Gamma_+$, and choose $L_m > 0$ such that

$$L_m > \sup_{x,y \in U_m} \|x - y\|_\infty.$$

For

$$\mathbb{T}_m^N := \mathbb{R}^N / L_m \mathbb{Z}^N,$$

the quotient map restricts to a diffeomorphism

$$\kappa_m : U_m \rightarrow \Omega_m := \kappa_m(U_m),$$

which is a local isometry. Since the Schwartz kernel of $A_{f,m}$ is supported in $K_m \times K_m$, with respect to

$$L^2(\Gamma; \mathbb{C}^G) = L^2(U_m; \mathbb{C}^G) \oplus L^2(\Gamma \setminus U_m; \mathbb{C}^G)$$

we have

$$A_{f,m} = B_{f,m} \oplus 0$$

for some $B_{f,m}$ on $L^2(U_m; \mathbb{C}^G)$.

Let

$$W_m : L^2(U_m; \mathbb{C}^G) \rightarrow L^2(\Omega_m; \mathbb{C}^G), \quad W_m u := u \circ \kappa_m^{-1}.$$

Then W_m is unitary. By the invariance of classical pseudodifferential operators under smooth coordinate changes [?, Theorem 18.1.17],

$$T_{f,m} := W_m B_{f,m} W_m^{-1} \in \Psi_{\text{cl}}^{-1}(\Omega_m; \mathbb{C}^G).$$

Its Schwartz kernel is supported in $\kappa_m(K_m) \times \kappa_m(K_m) \Subset \Omega_m \times \Omega_m$. Choose $\vartheta_m \in C_c^\infty(\Omega_m)$ equal to 1 on a neighborhood of $\kappa_m(K_m)$. Then

$$T_{f,m} = M_{\vartheta_m} T_{f,m} M_{\vartheta_m}.$$

Hence the zero extension of the Schwartz kernel of $T_{f,m}$ from $\Omega_m \times \Omega_m$ to $\mathbb{T}_m^N \times \mathbb{T}_m^N$ is a classical pseudodifferential kernel of order -1 : locally in $\Omega_m \times \Omega_m$ this follows from $T_{f,m} \in \Psi_{\text{cl}}^{-1}$, while near the boundary of $\Omega_m \times \Omega_m$ the kernel vanishes. Denote the corresponding operator on $\mathbb{T}_m^N$ by $\hat{A}_{f,m}$. Thus,

$$(32) \quad \hat{A}_{f,m} \in \Psi_{\text{cl}}^{-1}(\mathbb{T}_m^N; \mathbb{C}^G).$$

Since κ_m is a local isometry, by (31) and the transformation law for principal symbols,

$$(33) \quad \sigma_{-1}(\hat{A}_{f,m})(\kappa_m(x), \xi) = \frac{\chi_m(x)^2}{\|\xi\|} \mathbf{c}_{f,j} \left(x, \frac{\xi}{\|\xi\|} \right), \quad x \in U_m, \quad \xi \neq 0,$$

and the principal symbol vanishes outside Ω_m .

With respect to

$$L^2(\mathbb{T}_m^N; \mathbb{C}^G) = L^2(\Omega_m; \mathbb{C}^G) \oplus L^2(\mathbb{T}_m^N \setminus \Omega_m; \mathbb{C}^G),$$

we have

$$\widehat{A}_{f,m} = T_{f,m} \oplus 0.$$

Since $T_{f,m}$ and $B_{f,m}$ are unitarily equivalent,

$$(34) \quad \mu(t, \widehat{A}_{f,m}) = \mu(t, A_{f,m}), \quad t > 0.$$

By (32) and the Weyl law [?, Theorem 6.1],

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, \widehat{A}_{f,m}) = \left(\frac{1}{N(2\pi)^N} \int_{S^* \mathbb{T}_m^N} \text{tr}_{\mathbb{C}^G} |\sigma_{-1}(\widehat{A}_{f,m})(z, \theta)|^N dz d\theta \right)^{1/N}.$$

Since the metric on $\mathbb{T}_m^N$ is flat, combining (33) with (34), we obtain

$$(35) \quad \lim_{t \rightarrow \infty} t^{1/N} \mu(t, A_{f,m}) = \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \chi_m(x)^{2N} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N},$$

where χ_m is identified with its zero extension to Γ .

By Lemma 12.4,

$$(x, \theta) \mapsto \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N$$

belongs to $L^1(\Gamma \times \mathbb{S}^{N-1})$. Since $0 \leq \chi_m \leq 1$, $\chi_m \rightarrow 1$ on Γ_+ , and $\Gamma \setminus \Gamma_+$ is contained in finitely many hyperplanes, the dominated convergence theorem implies

$$(36) \quad \begin{aligned} & \lim_{m \rightarrow \infty} \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \chi_m(x)^{2N} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \\ &= \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}. \end{aligned}$$

By (30), $A_{f,m} \rightarrow A_f$ in $S^{N,\infty}$. Combining (35), (36), and Lemma 12.6, we obtain

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, A_f) = \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathbf{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Since U and V are unitary,

$$\mu(t, A_f) = \mu(t, [R_{\text{Dunkl},j}, M_f]), \quad t > 0.$$

This completes the proof of Lemma 12.5. $\square$

Lemma 12.6. *Let $0 < p < \infty$, let $\mathcal{H}$ be a Hilbert space, and let $A, A_m \in S^{p,\infty}(\mathcal{H})$, $m \geq 1$, satisfy*

$$\|A_m - A\|_{S^{p,\infty}} \rightarrow 0.$$

Suppose that, for every $m \geq 1$, the limit

$$c_m := \lim_{t \rightarrow \infty} t^{1/p} \mu(t, A_m)$$

exists. Then both limits

$$\lim_{m \rightarrow \infty} c_m \quad \text{and} \quad \lim_{t \rightarrow \infty} t^{1/p} \mu(t, A)$$

exist and

$$\lim_{t \rightarrow \infty} t^{1/p} \mu(t, A) = \lim_{m \rightarrow \infty} c_m.$$

Proof. The assertion is the Birman–Solomyak approximation lemma; see [?, Section 11.6]. $\square$

Lemma 12.7. *Assume that $N > 1$, $j \in \{1, \dots, N\}$, and $f \in \dot{W}^{1,N}(\mathbb{R}^N)$. Then there exists $f_m \in C_c^\infty(\mathbb{R}^N)$ such that*

$$\begin{aligned} \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} &\longrightarrow 0, \\ \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} &\longrightarrow 0, \\ \|\mathfrak{c}_{f_m,j} - \mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} &\longrightarrow 0. \end{aligned}$$

Proof. By the density of $C_c^\infty(\mathbb{R}^N)$ in $\dot{W}^{1,N}(\mathbb{R}^N)$, choose $f_m \in C_c^\infty(\mathbb{R}^N)$ such that

$$\|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \longrightarrow 0.$$

By Theorem ?? and Lemma ??,

$$\begin{aligned} \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} &= \|[R_{\text{Dunkl},j}, M_{f_m-f}]\|_{S^{N,\infty}} \\ &\lesssim \|f_m - f\|_{OSC^{N,\infty}(\mathbb{R}^N)} \lesssim \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \longrightarrow 0. \end{aligned}$$

By the linearity of $f \mapsto \mathfrak{c}_{f,j}$ and Lemma 12.4,

$$\begin{aligned} \|\mathfrak{c}_{f_m,j} - \mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} &= \|\mathfrak{c}_{f_m-f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \\ &\lesssim_{N,R,k} \|\nabla(f_m - f)\|_{L^N(\mathbb{R}^N, dx)} \longrightarrow 0. \end{aligned}$$

This completes the proof of Lemma 12.7. $\square$

Theorem 12.8. *Assume that $N > 1$ and $j \in \{1, \dots, N\}$. Then, for every $f \in \dot{W}^{1,N}(\mathbb{R}^N)$,*

(37)

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_f]) = \left(\frac{1}{N(2\pi)^N} \int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N}.$$

Moreover,

$$(38) \quad \left(\int_{\Gamma} \int_{\mathbb{S}^{N-1}} \text{tr}_{\mathbb{C}^G} |\mathfrak{c}_{f,j}(x, \theta)|^N d\theta dx \right)^{1/N} \simeq_{N,R,k} \|\nabla f\|_{L^N(\mathbb{R}^N, dx)}.$$

Proof. By Lemma 12.7, choose $f_m \in C_c^\infty(\mathbb{R}^N)$ such that

$$(39) \quad \|[R_{\text{Dunkl},j}, M_{f_m}] - [R_{\text{Dunkl},j}, M_f]\|_{S^{N,\infty}} \longrightarrow 0,$$

$$(40) \quad \|\mathfrak{c}_{f_m,j} - \mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \longrightarrow 0.$$

By Lemma 12.5, for every m ,

$$\lim_{t \rightarrow \infty} t^{1/N} \mu(t, [R_{\text{Dunkl},j}, M_{f_m}]) = \frac{1}{(N(2\pi)^N)^{1/N}} \|\mathfrak{c}_{f_m,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))}.$$

By (40),

$$\|\mathfrak{c}_{f_m,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))} \longrightarrow \|\mathfrak{c}_{f,j}\|_{L^N(\Gamma \times \mathbb{S}^{N-1}; S^N(\mathbb{C}^G))}.$$

Combining this convergence with (39), we apply Lemma 12.6 with $p = N$ and obtain (37). Finally, (38) follows from Lemma 12.4. This completes the proof of Theorem 12.8. $\square$

APPENDIX A. ZHIJIE LEMMA

Lemma A.1. *Let $1 < s < \infty$, and let $\mathbb{D}^0$ be the standard dyadic grid in $\mathbb{R}^N$. Suppose that $F \in L^s(\mathbb{R}^N)$. Then there exists a constant $C > 0$ such that*

$$\left\| \left\{ |Q|^{-1/s'} \int_Q |F(x)| dx \right\}_{Q \in \mathbb{D}^0} \right\|_{\ell^{s,\infty}} \leq C \|F\|_{L^s(\mathbb{R}^N)}.$$

Proof. Let $\mathcal{E} \subset \mathbb{D}^0$ be finite and set

$$G_{\mathcal{E}}(x) := \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Since the dyadic cubes containing any fixed x are totally ordered by inclusion, for each $x \in \bigcup_{Q \in \mathcal{E}} Q$ we may choose a cube $Q_x \in \mathcal{E}$ of minimal volume such that $x \in Q_x$. Every $Q \in \mathcal{E}$ containing x is then a dyadic ancestor of Q_x . Since the volume increases by the factor 2^N at each generation,

$$G_{\mathcal{E}}(x) = \sum_{\substack{Q \in \mathcal{E} \\ x \in Q}} |Q|^{-1/s'} \leq |Q_x|^{-1/s'} \sum_{k=0}^{\infty} 2^{-kN/s'} \lesssim_{N,s} \sup_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

Consequently,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)}^{s'} \lesssim_{N,s} \int_{\mathbb{R}^N} \sup_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx \leq \int_{\mathbb{R}^N} \sum_{Q \in \mathcal{E}} |Q|^{-1} \chi_Q(x) dx = \#\mathcal{E}.$$

and hence,

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \lesssim_{N,s} (\#\mathcal{E})^{1/s'}.$$

By Hölder's inequality,

$$\sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx = \int_{\mathbb{R}^N} |F(x)| G_{\mathcal{E}}(x) dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.$$

Fix $t > 0$ and let $\mathcal{E}$ be any finite subcollection of the dyadic cubes satisfying

$$|Q|^{-1/s'} \int_Q |F(x)| dx > t.$$

Then

$$t \#\mathcal{E} < \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)| dx \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'},$$

so

$$\#\mathcal{E} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s.$$

Since $\mathcal{E}$ is arbitrary,

$$\#\left\{ Q \in \mathbb{D}^0 : |Q|^{-1/s'} \int_Q |F(x)| dx > t \right\} \lesssim_{N,s} t^{-s} \|F\|_{L^s(\mathbb{R}^N)}^s, \quad t > 0.$$

The asserted weak ℓ^s estimate follows. This completes the proof of Lemma A.1. $\square$

REFERENCES

- [1] Nicolas Bourbaki. *Lie groups and Lie algebras. Chapters 4–6*. Elements of Mathematics (Berlin). Springer-Verlag, Berlin, 2002. Translated from the 1968 French original by Andrew Pressley. [3](#)
- [2] Jacek Dziubański and Agnieszka Hejna. Remark on atomic decompositions for the Hardy space H^1 in the rational Dunkl setting. *Studia Math.*, 251(1):89–110, 2020. [9](#), [32](#)
- [3] Z. Fan, M. Lacey, and J. Li. Schatten classes and commutators of Riesz transform on Heisenberg group and applications. *J. Fourier Anal. Appl.*, 29(2):Paper No. 17, 28, 2023. [3](#)
- [4] Z. Fan, M. Lacey, J. Li, M.N. Vempati, and B.D. Wick. Besov space, Schatten classes, and commutators of Riesz transforms associated with the Neumann Laplacian. *Michigan Math. J.*, 75(1):199–224, 2025. [3](#)
- [5] Z. Fan, M.T. Lacey, J. Li, and X. Xiong. Schatten–Lorentz characterization of Riesz transform commutator associated with Bessel operators. Preprint available at <https://arxiv.org/abs/2403.08249>, 2024. [3](#)
- [6] Y. Han, M. Lee, J. Li, and B.D. Wick. Riesz transforms and commutators in the Dunkl setting. *Anal. Math. Phys.*, 14(3):Paper No. 46, 32, 2024. [13](#)
- [7] G. Heckman. Coxeter groups. Lecture notes available at <https://www.math.ru.nl/~heckman/CoxeterGroups.pdf>. Radboud University Nijmegen, 2018. [29](#)
- [8] S. Helgason. *Differential geometry, Lie groups, and symmetric spaces*, volume 80 of *Pure and Applied Mathematics*. Academic Press, Inc. [Harcourt Brace Jovanovich, Publishers], New York-London, 1978. [2](#)
- [9] T. Hytonen. A_∞ -invariance of oscillatory norms and schatten characterisations of commutators. [17](#)
- [10] T. Hytonen. Schatten properties of commutators on metric spaces. Preprint available at <https://arxiv.org/abs/2411.02613>, 2024. [3](#), [4](#), [17](#)
- [11] M.T. Lacey, J. Li, and B.D. Wick. Schatten classes and commutator in the two weight setting, I. Hilbert transform. *Potential Anal.*, 60(2):875–894, 2024. [3](#)
- [12] J. Li, X. Xiong, and F. Yang. Schatten properties of Calderón–Zygmund singular integral commutator on stratified Lie groups. *J. Math. Pures Appl. (9)*, 188:73–113, 2024. [3](#)
- [13] R. Rochberg and S. Semmes. Nearly weakly orthonormal sequences, singular value estimates, and Calderon–Zygmund operators. *J. Funct. Anal.*, 86(2):237–306, 1989. [3](#)
- [14] Z. Wei and H. Zhang. Complex median method and Schatten class membership of commutators. Preprint available at <https://arxiv.org/abs/2411.05810>, 2024. [3](#)
- [15] Z. Wei and H. Zhang. On the boundedness and Schatten class property of noncommutative martingale paraproducts and operator-valued commutators. Preprint available at <https://arxiv.org/abs/2407.14988>, 2024. [3](#)

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